

A combined data mining – optimization approach to manage trucks operations in container terminals with the use of a TAS: Application to an Italian and a Mexican port

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ABSTRACT

This paper relates with the assignment of trucks to time slots in container terminals equipped with Truck Appointment Systems. A two-phase approach is provided: first, export and import containers are matched in tuples with a clustering analysis to reduce the number of empty trips and, then, tuples are assigned to time slots to minimize trucks deviation from their preferred time slots and truck turnaround times. Real case instances related to Mexican and Italian container terminals are tested. Results show that our approach reduces empty-truck trips up to 33.79% and that it can be successfully applied to any container terminal.

1. Introduction

In the last decades, the dynamics generated by international maritime trade and naval gigantism have changed the pattern of container flows handled in ports. An increasing quantity of freight, to be handled in short time windows, imposes container terminals to enhance their planning capabilities and efficiency. Since a high percentage of containers are transferred to and from terminals by road transport, terminals must manage, in small time intervals, large quantities of trucks that have to pick up and deliver containers. The fluctuation and unpredictability of truck arrivals exacerbate this phenomenon, creating peaks that may easily exceed the terminal processing capabilities. These peaks strongly impact the terminal efficiency and productivity, causing severe congestion problems that, if not properly managed, may affect the whole logistic network.

To mitigate these negative impacts, two main solutions may be implemented: to reduce the number of trucks visiting the terminal, or to improve the management and control of truck flows. The first option can be pursued by effectively exploit trucks' capacity in terms of the number of containers to be simultaneously loaded. As it is well known, trucks can transfer a maximum of two containers at a time. A *double move* is executed when, in the same trip, a truck arrives at a container terminal to both deliver one or two *export* containers, and then pick up one or two *import* containers to be delivered in the hinterland. Double moves imply that trucks are never empty.

Regarding the second aspect, one of the most promising and quite recent solutions is the application of Truck Appointment Systems (TASs), i.e., information systems –usually managed by terminals– which allow truckers to book their arrival at the terminal in a specific time slot of the gate opening time windows. These systems can be compulsory or optional. In the latter case the terminal

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allows truckers to arrive without a booking but, at the same time, does not guarantee to serve it within a predetermined service level, that is usually guaranteed for booked trucks.

Some ports have already implemented them (Giuliano and O'Brien, 2007; Huynh and Walton, 2008; Li et al., 2018), aiming to modify the current arrival patterns of external trucks, adjusting them in accordance with the resources available in the terminals. However, a typical TAS only partially solves the problem since many trucks arrive at a terminal to deliver or pick up only one container on a selected time slot (i.e., no double moves are performed). The best operation for trucks, then, consists in reaching the terminal with two containers to be delivered and leaving the terminal again with two containers. As will be further detailed (with reference to some real case studies), this is not the most common way in which trucks operate; nonetheless, along this reasoning line, the double moves are already pursued by truckers.

So, the possibility of both reducing the number of trucks (exploiting the full double move operativity) and managing the arrival times of trucks (by using TASs) appears to be the most effective way to reduce emissions, congestion and capacity problems for major container terminals (Huynh and Zumerchik, 2010).

In order to address this problem, we propose a new methodological approach that combines operational research (OR) and data mining (DM) techniques to properly match import containers with arriving trucks that have to perform export deliveries within specific time slots, so to drastically reduce empty truck trips and, therefore, also minimize congestion inside the terminal by better distributing the workload over gate opening hours. Empty truck trips, caused by imbalanced traffic flows, bad planning capabilities, and/or lack of coordination among truckers, are responsible for a decrease in truck carriers' profit margins, congestion, and a significant amount of port-related emissions.

The synergies between OR and DM techniques have been studied in Olafsson et al. (2008), Meisel and Mattfeld (2010), and Corne et al. (2012). However, to the best of our knowledge, the approach of combining OR and DM techniques has been never adopted in the context of drayage operations in ports. In our methodological framework, DM techniques are used to analyze the vast amount of data –that are daily collected by terminals– in order to create proper clusters of export–import containers, whereas OR models are used to obtain optimal solutions from a variant of the assignment problem.

This new methodological approach is very effective. First of all, it allows to significantly reduce the complexity of the optimization problem, since the creation of tuples is delegated to data mining techniques. Moreover, it is more versatile and flexible than similar methodologies proposed in the literature since it allows to easily change container attributes in the matching phase without the necessity of modifying the mathematical model.

Two experimental frameworks have been considered and analyzed: (1) an experimental analysis (i.e., a Design of Experiments - DOE) using a full randomized factorial design study with the aim of identifying the main factors affecting the response variables, and (2) the application of this methodology to two real case studies (i.e., the container terminal ATP in the Mexican port of Altamira and the container terminal of PSA – Genova Pra located in the Italian port of Genoa) with the objective of validating the proposed methodology.

As shown by the results obtained by applying the proposed methodology to two very different size container terminals, the approach is general and robust enough to be applied to any container terminal located anywhere worldwide.

In the following Section, the differences between our mathematical model and those already presented in the literature will be highlighted.

1.1. Related literature

Compared to other works present in the literature, the foremost contribution of this study is the proposal of a new methodology that combines data mining (DM) –for the implementation of a clustering analysis– and mathematical optimization (OR) techniques to jointly solve the combined empty truck trips and appointment assignment problems, and on the application of this new method to two real container terminals located in different countries. More specifically, the goal of the clustering analysis (whose output is provided as input to the mathematical programming model) is to increase double moves, that means minimize the number of empty truck trips, by properly matching containers on the basis of common features (type, weight, dwell time, final destination, customs clearance, etc.). The objective of the mathematical model is, instead, to firstly minimize the deviation between the time slots assigned to trucks by the TAS and their preferred time slots and, secondly, to minimize truck turnaround times (that also means to increase truck service levels).

In order to better highlight the contribution of our work, we refer the reader to Table 1 which shows features, objectives and methodologies of the papers already present in the literature that have faced the study of a TAS. As it can be noted in Table 1, we have identified three main decisions regarding the design of a TAS: (A) time windows length definition, (B) appointment quota definition, and (C) truck arrivals management (subdivided in C1-truck arrivals scheduling, C2-deviation from preferred time of arrival, and C3-collaboration/truck-sharing agreements). The first decision (A) is explicitly tackled by Chen and Yang (2010) and Phan and Kim (2016); the definition of the appointment quota (B) is faced by Huynh and Walton (2011), Zhang et al. (2019), Chen et al. (2011), Zehendner and Feillet (2014) and Li et al. (2019), while our work and many others (Chen et al., 2011, 2013; Chen and Yang, 2010; Corne et al., 2012; Huynh and Walton, 2008, 2011; Huynh and Zumerchik, 2010; Islam et al., 2013; Li et al., 2018, 2019; Meisel and Mattfeld, 2010; Olafsson et al., 2008; Phan and Kim, 2015, 2016; Zehendner and Feillet, 2014; Zhang et al., 2019; Zouhaier et al., 2017) deal with the third decisional aspect (C); however, only (Phan and Kim, 2016; Phan and Kim, 2015) and our paper consider all the aspects related to truck arrival management (i.e., C1, C2 and C3).

The fulfillment of the desired truck arrival time (column C2 of Table 1) is a critical aspect affecting truckers service level and satisfaction and, therefore, some authors have already addressed it. For instance, Li et al. (2019) propose a bi-objective model to

Table 1
Literature review summary

No.	Author	Year	TAS					Goals			Approach	
			A	B	C			D	E	F	G	H
					C1	C2	C3					
1	Giuliano and O'Brien (2007)	2007			Truck emissions reduction				✓	✓		
2	Huynh and Walton (2008)	2008			✓			✓	✓			✓
3	Li et al. (2018)	2018			✓	✓		✓	✓			
4	Chen and Yang (2010)	2010	✓					✓	✓			✓
5	Phan and Kim (2016)	2016	✓		✓	✓	✓	✓	✓			✓
6	Huynh and Walton (2011)	2011		✓				✓				
7	Zhang et al. (2019)	2019		✓		✓		✓	✓			✓
8	Chen et al. (2011)	2011		✓	✓	✓		✓	✓			✓
9	Zehendner and Feillet (2014)	2014		✓		✓		✓	✓			✓
10	Li et al. (2019)	2019		✓				✓	✓			✓
11	Islam et al. (2013)	2013			✓		✓		✓	✓		
12	Chen et al. (2013)	2013			✓	✓		✓				✓
13	Phan and Kim (2015)	2015			✓	✓	✓		✓			✓
14	Zouhaier et al. (2017)	2017			✓	✓			✓			✓
15	Schulte et al. (2017)	2017			✓		✓	✓	✓	✓		✓
16	Dekker et al. (2013)	2013			Chassis exchange terminals			✓	✓			
17	van Asperen et al. (2013)	2013			Impact of TAS on stacking			✓				
18	Ambrosino and Caballini (2015)	2015			Congestion reduction			✓	✓			
19	Caballini et al. (2018)	2018			✓			✓	✓			✓
	This Paper				✓	✓	✓	✓	✓	✓		✓

Note. *TAS features:* A– Definition of time windows length, B– Appointment quota sizing, and C– Truck arrivals management. Where C1– Truck arrivals scheduling, C2– deviation from preferred time of arrival, C3– collaboration or truck-sharing agreements. *Goals:* D– Reduce truck turnaround times and/or waiting times inside the terminals, E– Reduce congestion, F– Reduce empty truck trips. *Methodology:* G– Optimization problem, H– Data mining.

determine the appointment quota with the final aim of balancing the trade-off between terminals (to minimize the number of yard cranes) and trucks (to minimize the average waiting time). Zehendner and Feillet (2014) propose a MILP model to determine the number of appointments (appointment quota) to match the preferred arrival times of trucks by allocating straddle carriers to different transport modes with the aim of minimizing the overall delays at the terminal. Chen et al. (2011) use an analytical point-wise stationary approximation model to analyze time-dependent truck queuing processes with stochastic service time distributions at gates and yards of a port terminal; they propose a two phase optimization approach to compute a system-optimal truck arrival pattern in the first phase, and then find a desirable pattern of time-varying tolls that leads to the optimal arrival pattern in the second phase. Note that Chen et al. (2011) considers aggregate truck arrival patterns whereas our model takes into account each truck separately, in a disaggregate way.

The implementation of a TAS leads to the problem of coordinating various actors involved in the port cycle. Therefore, collaboration among truckers (column C3 of Table 1) is seen as the major opportunity to address this issue, which would reduce costs, congestion and emissions. Phan and Kim (2015) suggest an appointment process by which trucking companies and terminals collaboratively determine truck operation schedules and truck arrival appointments. The same authors (Phan and Kim, 2016) have later proposed an iterative collaboration process between trucking companies and terminals to collaboratively determine truck operation schedules and truck arrival appointments. Schulte et al. (2017) propose a graph-based collaborative mathematical model for a truck scheduling problem with collaboration, to be operated within a TAS, and they investigate its impact on emissions and cost objectives.

Other papers concentrate on other aspects such as truck emission reductions (Giuliano and O'Brien, 2007), chassis exchange terminals (Dekker et al., 2013), TAS impacts on yard stacking operations (van Asperen et al., 2013) or explicitly focus on the reduction of congestion (Ambrosino and Caballini, 2015). In more detail, Giuliano and O'Brien (2007) evaluate the impacts produced by a TAS in terms of truck emissions and congestion reduction, applied to the port of Los Angeles and Long Beach in California. The study of Dekker et al. (2013) suggests the chassis exchange terminal concept to increase the utilization of container truck capacity and to minimize the number of empty-truck trips. van Asperen et al. (2013) evaluate the impact of a truck announcement system on the performance of online container stacking rules, whereas Caballini et al. (2018) propose a mathematical model to assign appointments to the terminal time slots with the goals of optimizing terminal resources, guaranteeing a predefined service level to haulers and reducing congestion inside the terminal.

Among the considered goals, the majority of the papers in the literature try to reduce truck turnaround times and/or truck waiting times (column D of Table 1), while some others try to reduce congestion (column E of Table 1). Note that, within Column D, we identify papers that seek to reduce the time needed by a truck to travel from the gate-into the gate-out of the terminal (i.e., the truck turnaround time), so including only eventual congestion inside the terminal. In contrast, on Column E we consider the papers where congestion goals are addressed from a broader perspective, both inside and outside the terminal. Only a minority of the literature papers explicitly deals with the reduction of empty truck trips (column F of Table 1), even if this aspect is crucial to reduce congestion

and environmental issues as well as increase profit margins for trucking companies.

The empty-truck trip problem is rooted in the imbalance of trade flows (Islam et al., 2013) and is triggered by the lack of collaboration among supply chain partners (especially among shippers, freight forwarders and trucking companies), causing inefficient use of trucks capacity and severe environmental issues. The empty truck trips problem has been approached in many disciplines (Dejax and Crainic, 1987); however, research covering the empty truck trip problem in the port logistic domain is still scarce. Only few authors, such as Schulte et al. (2017), have addressed it.

Finally, regarding the proposed methodology, the majority of the papers use optimization techniques while some of them use simulation (such as Dekker et al., 2013); however, none of them use OR in combination with DM techniques nor consider the above mentioned aspects all together. This paper tries to fill this gap. Besides, the idea of considering tuples (i.e., association of containers) is novel in the context of container transportation.

The remainder of this paper is organized as follows. Section 2 describes in detail the problem under investigation. In Sections 3 the proposed methodology is described, while in Section 4 an extensive DOE experimental campaign is performed. Section 5 describes the application of the proposed approach to two different real case studies: the PSA – Genova Pra container terminal located in the Italian port of Genoa and the Altamira Container Terminal in Mexico. Finally, Section 6 reports some concluding remarks.

2. Problem description

Let us consider a typical import and export process of any port system and concentrate on the hauling cycle, i.e., the travel between the shipper (exporter, importer or freight forwarder warehouse) and the container terminal and/or the other way around. In case the terminal is provided with a TAS, the terminal operator determines the appointment quota for each time slot (i.e., the number of containers that can be handled per time slot) trying to satisfy some performances (e.g., maximize its resource productivity, minimize truck service time, reduce gate congestion or maximize the number of trucks handled by the terminal). The available appointments are then published on the terminal website/application (or other on line tools such as the PCS-Port Community System) to be available for trucking companies. Thus, each truck may select one time slot to perform the delivery or pick up of its container(s). To do so, the truck operator has to browse, with some advance, the terminal website and book its arrival by providing all the required information (ID container, company, truck plate, trucker ID, booking number, etc.), selecting its preferred time slot from the TAS. If the preferred time slot is fully booked, the trucker will have to choose another one. Once the slot is booked, the trucker can arrive at the container terminal gate to deliver or pick up its container. Note that this booking process can be avoided in case the TAS is not compulsory. In that case, the trucker may show up at the terminal without making an appointment and, usually, it will encounter a longer turnaround time since it will be served only when the terminal resources are available without a guaranteed service level.

Fig. 1 provides a general sketch of the considered problem, where the double move process in a generic container terminal is shown: *truck 2* arrives at the terminal gate with two export containers to deliver, colored respectively in blue and orange colors (step 1). Each container has to be released in a specific area of the yard (shown in Fig. 1 with a specific color). Once executed the gate-in (i.e., all the activities related to the entrance of the truck in the terminal), the truck can enter the terminal to release its containers: first it has to reach *area 6* to release the blue container (step 2) and, then, it goes to *area 9* to leave the orange one (step 3). Once the export cycle has been performed, the truck can travel to the import areas to collect its import container(s). In this case two import containers have to be picked up: the blue one has to be collected in *area 3* (step 4), while the pink container is stacked in *area 1* (step 5). Once these operations have been performed, the truck can execute the terminal gate-out and exit the terminal (step 6) to travel towards the hinterland in order to bring its import containers at their final destinations.

Note that each terminal equipment (as RMGs cranes used in Fig. 1) is characterized by a different productivity (movements per

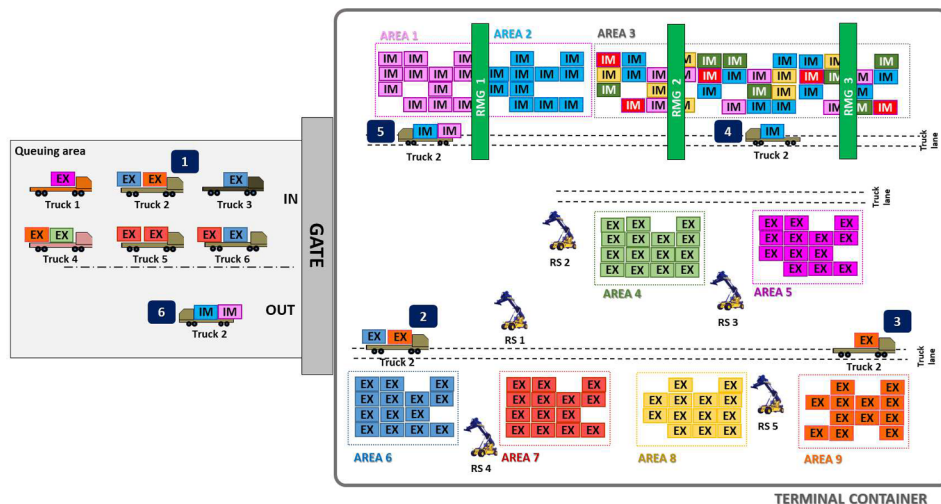


Fig. 1. A sketch of the double move truck cycle.

Table 2
Possible cases of container pick up and delivery in a container terminal.

Case	# of export cntrs	# of import cntrs	Total # of cntrs
1	1	0	1
2	2	0	2
3	0	1	1
4	0	2	2
5	1	1	2
6	2	1	3
7	1	2	3
8	2	2	4

time unit); this productivity depends on the particular type of equipment and on the level of filling/congestion of the terminal area (which directly affects the number of rehandles, i.e., unproductive movements to be performed) where the equipment is working in a particular time slot. Even if Fig. 1 proposes two different kinds of terminal equipment (i.e., RMG cranes and reach stackers), it has to be pointed out that the mathematical problem presented in this paper is general enough to consider any type of terminal handling mean.

As shown in Table 2, a generic truck arriving at a container terminal may perform from one to four operations; therefore, eight different cases may arise.

Considering the magnitude of truck arrivals in medium to large sized container terminal, the number of empty truck trips generated may be quite relevant, so creating congestion issues and increasing environmental impacts (air emissions, noise, etc.). The increase of double moves can significantly address this problem.

Formally speaking, the optimal scheduling of trucks operations in truck appointment systems can be defined as follows: *given a set of trucks that book an appointment through the terminal TAS by providing information related to their preferred time slot, as well as the number and features of export and import containers to be delivered or picked up, determine the optimal assignment of appointments.* The related goals are usually one or all the following ones: to optimize the truck service level, minimize terminal congestion, reduce empty truck trips and/or maximize the use of terminal resources.

For example, if a truck prefers to arrive at the terminal at 8:00 a.m., the model may indicate that its determined time slot would be at 9:30 a.m., as in this time slot the truck turnaround time would be minimized, the container terminal resources would be optimized and one or two import containers would be available to be picked-up to avoid an empty trip.

The general problem considered in this work could be stated as a mathematical programming problem with several objective functions and, in particular, it would be a complex Mixed Integer Linear Programming (MILP) problem. Instead, the foundation of our approach consists in coupling data-driven clustering process and a simpler assignment problem, with the benefit of reducing the complexity and size of the problem. We refer the reader to Appendix A where the more complex formulation of the problem is presented.

More specifically, data mining techniques are properly used to define a set of *tuples*. A tuple is composed of the following information {truck, export container 1, export container 2, import container 1, import container 2}. Then, a tuple is the best set of loading units to be transported by a truck depending on the original information about the truck and the various containers to be handled. Once the set of tuples is created, it is possible to assign each tuple (i.e., each truck) to a specific time slot. It is important to note that, in the real context, trucks are already associated with the containers they have to move (in fact this decision is taken by the subject that is in charge of the road transportation planning, usually a freight forwarder or the shipper), while in this work we propose to change philosophy and to decide what a truck should transport.

The following assumptions are considered:

- import and export containers are assigned to different yard areas, to facilitate delivering and pick up processes. This hypothesis is rather realistic since the majority of terminals operate in this way;
- each time slot has a time length equal to 1 h;
- from one to four containers can be associated with each truck, labelled *exp1* and *exp2*, *imp1* and *imp2*, respectively, whose delivery or pick-up areas (*area1*, *area2*, *area3*, *area4*) are known;
- the container terminal knows the preferred time slot of each truck/tuple. This preferred time slot corresponds to the time slot previously selected by the truck when making the booking through the TAS application;
- the container terminal knows the information of each container such as size, weight, customs clearance, release time, yard location and destination in the hinterland.

3. The adopted methodology

To address the problem sketched above, a two-phase hierarchical approach that combines a data mining technique with a mathematical programming model is proposed. Fig. 2 provides a scheme of the methodology. In phase 1, all the available information regarding import and export containers is used to create disjoint sets of tuples. The resulting tuples are then used as input to the

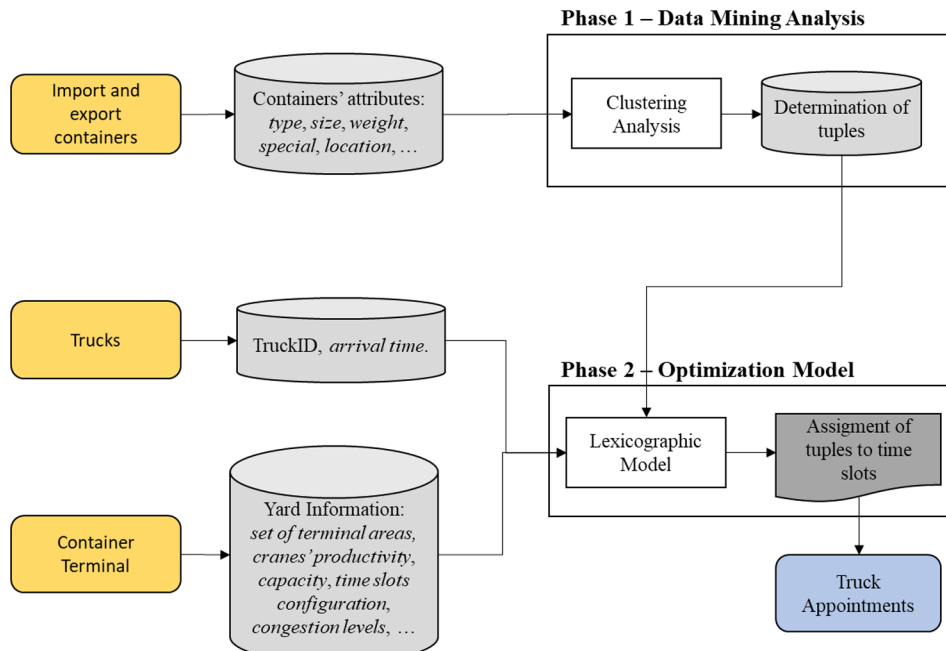


Fig. 2. A scheme of the adopted methodology.

assignment problem (phase 2) which provides as output a detailed allocation of trucks to the terminal time slots.

The clustering analysis aims to create a feasible set of tuples, i.e., groups of containers that may be transported together. As a result of clustering import and export containers together, it is possible to: (1) reduce the number of empty truck trips, (2) increase the number of double moves, (3) reduce the number of trucks needed to perform the pick-up and delivering operations, and (4) maximize the truck usage (i.e., two 20' containers or one 40'). Moreover, on equal demand for trips, by combining containers, the number of trips performed is lower, reducing congestion issues and environmental pollution due to the presence of trucks on the road network.

On the other hand, the optimization model try to minimize the deviation of time slots assigned to trucks from their preferred time slots, and minimize truck turnaround times, in a lexicographical order:

- first: minimize the deviation between the preferred time slot of a truck and its assigned time slot;
- second: minimize truck turnaround times.

Summarizing, the problem is solved as follows: (1) apply the *clustering approach* in order to group export and import containers within tuples according to specific criteria and rules; (2) solve an optimization problem to assign the previously created tuples to the terminal time slots.

3.1. Phase 1: Data mining analysis to create tuples

This phase aims to group import and export containers considering some predefining criteria in order to create tuples of the form $\{truckID, exp1, exp2, imp1, imp2\}$. Three stages are required: in a first stage, we group export containers in pairs to create *export subsets* (although an export subset may contain only one export container). In a second stage, we group import containers in pairs to create *import subsets* (which also may contain only one or two import containers). Finally, in a third stage, we match the export subset (with one or two export containers), with an *import subset*, so creating a *complete tuple*.

To create the tuples, i.e., to adequately match export (i.e., trucks) and import containers, we have used the *clustering analysis*. Clustering is an important unsupervised learning data mining function used to identify groups of objects that are *similar*, considering several attributes on a dataset that describes objects. The aim of clustering analysis is to create clusters where data objects within the same cluster are similar to one other, but dissimilar to the objects in other clusters. Clusters are created based on similarity metrics or probability density models. Generally, clustering techniques are classified into: partitioning methods, hierarchical methods and grid based methods, among others. We refer the reader to Jain et al. (1999) and Xu and Wunsch (2005) for a detailed review of clustering techniques and applications.

The hierarchical clustering method is recommended for our particular problem, as it generates a series of models with cluster solutions from 1 (all cases in one cluster) to n (one cluster for each individual case). A hierarchical clustering algorithm follows three steps: (1) compute the similarity coefficients between object i and j , (2) link the clusters through a clustering procedure, and (3)

choose a solution by selecting the appropriate matching of objects.

The general procedure for determining clusters of similar objects is to measure the similarity or dissimilarity of two objects. A similarity coefficient measures the resemblance between two objects based on a set of K attributes or variables. For computing the similarity between containers i and j , we use the Gower's general coefficient of similarity [Gower, 1971](#):

$$S_{ij} = \frac{\sum_{k=1}^K s_{ijk}}{\sum_{k=1}^N \delta_{ijk}} \quad (1)$$

where, S_{ij} is the similarity coefficient between items i and j ($0 \leq S_{ij} \leq 1$); s_{ijk} is the comparison score between items i and j compared on character k (we refer the reader to [Gower \(1971\)](#) to analyze some ways of calculating this score for each type of variable); δ_{ijk} represents the possibility of making comparison between items i and j with character k ($\delta_{ijk} = 1$ if i and j may be compared regarding character k , 0 otherwise). Note that when $\delta_{ijk} = 0$, then s_{ijk} is conventionally set to 0.

This similarity coefficient is a well-known technique to consider dichotomous (binary), qualitative (categorical) and quantitative (numerical) variables. It is computed as the average score taking all the possible comparisons. It differs from other similarity coefficients like *Jaccardian* ones, which are mainly used to take into account the binary characteristics of an item set. We refer the reader to [Jackson et al. \(1989\)](#), [Islam and Sarker \(2000\)](#), and to [Yin and Yasuda \(2006\)](#) for a review of similarity coefficients methods.

The following variables or collection of attributes have been used to describe the object "container" and make comparisons to determine its similarity with other containers:

- (i) *cycle*: binary variable that distinguishes between import and export containers {import, export};
- (ii) *ISOtype*: categorical variable representing the characteristics of the container (e.g., size) {22G1, 42G1, 45G1, 42R1, others}. This attribute is relevant as the physical capacity of trucks must be considered in the creation of tuples. A truck can carry maximum two 20ft containers (e.g. 22G1), or one 40ft container (e.g. 42G1, 45G1) in the same cycle (import or export);
- (iii) *weight*: numerical variable representing the weight of a container. This attribute is important because there is a maximum weight to be respected on a truck. Let w_i and w_j be the weight of the container i and j , respectively. Thus, both containers are allowed to be joined if $w_i + w_j \leq TON$, where TON is the maximum weight that may be moved by a truck. For instance, in Mexico $TON = 30$ tons;
- (iv) *special*: categorical variable that captures the special features of a container according to its cargo {IMO, Reefer, Normal}. This is an important attribute that constrains the possibility of matching containers in terms of compatibility on the truck (for instance, a container with dangerous goods cannot be matched with another dangerous container if they belong to incompatible categories), and of stacking distance. As a matter of fact IMO and reefer containers are stored in different areas of the terminal, separated by the other types of containers;
- (v) *agreement*: categorical variable that captures the existence of collaboration agreements between shippers or freight forwarders to share trucks (in single or double cycles). All the containers belonging to shippers/freight forwarders having an agreement share a common code;
- (vi) *vessel departure time*: categorical variable used to match only export containers. This attribute is useful to identify the departure time of the vessel on which the export container will be loaded. Note that associated with a vessel departure time there is a *vessel delivery window* (also called *yard opening time window*) and a *maritime service* associated. Thus, this attribute ensures that only export containers belonging to the same maritime service can be matched together;
- (vii) *distance*: dichotomous variable that considers the distance to be covered by the truck when covering the distance between two container (either two export, two import, or one import and one export) in the terminal. It takes the value of 1 if the distance travelled from the last position of the truck to the position where the next container is located is $\leq h$ minutes. The value of h is a parameter defined by the user. This will ensure that a truck does not spend too much time from a task to another, and that the distance travelled is minimal. Note that the number of empty trips may decrease as the value of h increases;
- (viii) *customs clearance*: binary variable that identifies whether or not custom clearance procedures have been fulfilled for import containers. If the procedure has been done prior to the arrival of the truck, then this is classified as cleared. Otherwise, it is considered to be not cleared and, therefore, it cannot be released from the terminal. {Cleared, Not Cleared}.
- (ix) *dwel time*: numerical variable representing the time that an import container has remained in the terminal yard. This attribute is relevant since, on other equal conditions, import containers are matched if they have similar dwell time, giving preference to assign first containers having higher dwell times so to minimize demurrage costs (note that the terminal charges demurrage fees after a certain number of days of free charge);
- (x) *destination*: categorical variable used to identify the final destination area of the container in the hinterland. This attribute is important because two import containers may be transported together when they have the same final destination.

Note that attributes (viii, ix and x) are used to match exclusively import containers. Attributes (ii, iii, iv, v and vi) are used to group export containers into pairs, whereas attributes (ii, iii, iv, v, viii, ix and x) are used to group import containers into pairs. Attribute (vii) is instead used to join an export group with an import group, as better defined below.

The process of clustering items (i.e., containers) into homogeneous groups (i.e., tuples) depends on two factors: (1) the clustering procedure selected, and (2) the threshold value of similarity adopted, which represents the minimal admissible level of similarity within the members of a group. There are different types of clustering techniques in the literature including: partitioning clustering, grid-based clustering, density-based clustering, and hierarchical clustering, among others. The partitioning clustering subdivides the

data sets into a set of k groups. The grid-based clustering partitions the data space into a finite number of cells to form a grid structure and then forms clusters from the cells in the grid structure. The density-based clustering starts with a data point that has not been visited, then all points which are within a predefined ϵ distance are identified as neighborhood points, and become part of the cluster. This method does not require a preset number of clusters. The hierarchical clustering is an alternative approach to identify groups in the dataset. It neither requires to pre-specify the number of clusters to be generated.

Within the hierarchical clustering procedures, the Average Linkage Clustering (ALC) algorithm has the advantage of showing the greatest robustness regardless of similarity coefficients (Yin and Yasuda, 2006). Therefore this has been used as clustering procedure to group the containers to form tuples. In ALC, the similarity coefficient between two clusters is defined as the average of the similarity coefficients between all members of the two clusters. In general, any threshold cut level of similarity can produce valid aggregations. However, accordingly to Lorenzo-Seva and Ten Berge (2006), a threshold value ranging from 0.85 to 0.94 means that the two objects compared display fair similarity, while a value higher than 0.95 implies that they can be considered equal. The ALC algorithm works as follows:

- (1) Create the *export subset*:
 - (1.1) identify the set of export containers that cannot be matched together with any other export container due to the transporting restrictions, considering attributes (ii, iii, and iv);
 - (1.2) with the remaining export containers, compute the similarity coefficients considering attributes (ii, iii, iv, v, and vi);
 - (1.3) join the two most similar objects (containers) considering the threshold value of similarity θ .
- (2) Create the *import subset*:
 - (2.1) identify the set of import containers that cannot be matched together with any other import container due to the transporting restrictions, considering attributes (ii, iii, and iv);
 - (2.2) with the remaining import containers, compute the similarity coefficients considering attributes (ii, iii, iv, v, viii, ix, and x);
 - (2.3) join the two most similar objects (containers) considering the threshold value of similarity θ .
- (3) An export subset and an import subset are joined to form a *complete tuple*:
 - (3.1) compute the similarity coefficients considering attributes (v and vii);
 - (3.2) join the two most similar objects (subsets) considering the threshold value of similarity θ .

This procedure was coded on C programming language.

Note that tuples are formed taking into account, among other factors, the constraints imposed by the deadlines (in terms of days and hours) of the containers. Therefore, the truck chooses a preferred time slot within the associated tuple's feasible range. This feasible interval is determined by considering the overlap of all the available time windows of the containers that form the tuple.

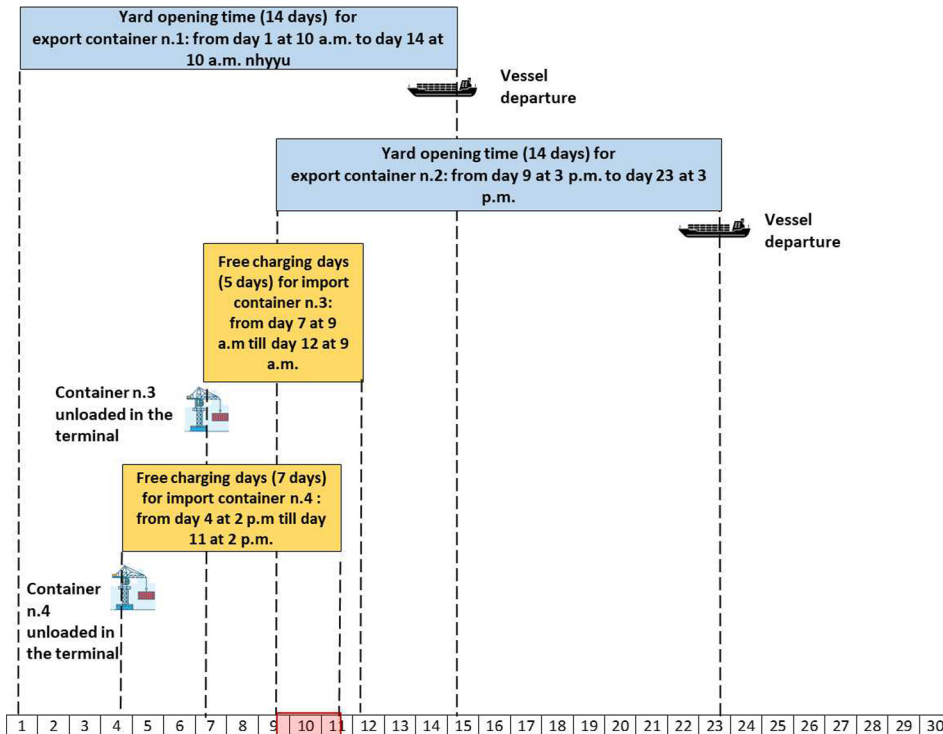


Fig. 3. An example of the preferred time slot chosen for a truck/tuple.

Fig. 3 shows an example of 4 containers matched in the same tuple: according to the different deadlines of the containers, the preferred time slot for the truck must be chosen in the feasible interval [day 9, 3 p.m. - day 11, 2 p.m.] since only in this interval all four container deadlines are met. Regarding the specific hour, the trucking company chooses, within the feasible interval, a preferred time slot that best fits its transportation planning objectives. Appendix B describes the rules used to define the preferred truck/tuple time slot.

At the end of the above explained procedure, each tuple (characterized by any of the eight cases defined in Table 2) is associated with a truck. The number of containers to be picked up or delivered by a truck (that will be provided by parameter u_n^a , as it will be better defined in the next subsection) is an indicator of how efficient is the clustering phase; in fact, good clusters result in trucks that have to visit few and as much as near areas.

3.2. Phase 2: Optimization model

In the second phase of the proposed approach, a mathematical programming model is solved to assign the tuples –created in the previous phase– to the time slots in which the gate opening time window is divided. Each truck and, hence, each tuple, is associated with a preferred time slot, which is the time a trucker would like to arrive at the terminal. However, since the terminal can guarantee only a maximum number of appointments in each time slot, it may happen that the preferred time slot cannot be fulfilled and the truck booking is shifted in advance or delay.

A mixed integer programming model is formulated to manage the operations of trucks that have to deliver and pick up containers in a container terminal operating with a TAS. More specifically, we have to decide in which time slot to serve the trucks approaching the terminal with the first goal of minimizing the deviation from their preferred time slot and, secondarily, minimize truck turn-around times: therefore, we propose a bi-objective optimization model that may be solved lexicographically. Although there are other goal programming approaches, in the context of well-defined preferences, the use of Lexicographic Goal Programming is recommended (Wilamowsky et al., 1990). The pursuing of the above mentioned goals will help, indirectly, to minimize terminal congestion and exploit at best terminal resources.

For modeling purposes, the following notation is adopted:

- S , is the set of time slots;
- I_s , is the time at which time slot $s = 1, \dots, S$ starts;
- F_s , is the time at which time slot $s = 1, \dots, S$ finishes;
- N , is the set of tuples obtained from the data mining phase;
- A , is the set of terminal areas;
- A_n , is the sequence of terminal areas visited by tuple $n \in N$, i.e., $A_n \subseteq A$ where $A_n = [a_1 \leq a_2 \leq \dots \leq a_4]$;
- u_n^a , is the number of containers (1 or 2) to be managed (i.e., to be delivered or picked up) for tuple $n \in N$ in area $a \in A_n$. Note that import and export containers are never stored in the same area since they are subject to different management policies and usually are handled by different type of equipment: this means that in a terminal area not more than 2 containers are ever picked up or delivered;
- α_n , $n \in N$, is a parameter representing the priority of a tuple. It takes the value of 1 if tuple n is performing a single cycle (i.e., only import or only export containers are moved: cases 1 to 4 of Table 2), whereas it takes the value of 2 if tuple n is performing a double cycle (cases 5 to 8 of Table 2). This means that we give priority to double moves in order to reduce empty trips. Of course, in order to express other preferences or policies, different values may be assigned to α_n for each of the eight move types of Table 2. Note that the higher the value of α_n , the higher the priority of tuple n to be assigned in a time slot, so reducing the deviation from its preferred time slot;
- g_n^s , is the preferred time slot at which the truck assigned to tuple $n \in N$ would like to arrive at the terminal for being served;
- δ_{max} , is the maximum value that δ_n can assume, so allowing higher or lower deviations from the preferred time slot of truckers;
- ρ_s^a , is the productivity of the yard handling resources (expressed in movements/time unit) assigned to area $a \in A$ in time slot $s = 1, \dots, S$;
- β_s^a , is the level of congestion in area $a \in A$ during time slot $s = 1, \dots, S$. Its value is set between 0.5 (very high congestion) and 1 (no congestion);
- t_n^{IN} , is the time needed for a truck belonging to tuple $n \in N$ to cover the distance from the terminal gate to the first area (i.e., the area where the truck has to perform its first operation);
- $t_n^{a,a+1}$, is the time needed for a truck belonging to tuple $n \in N$ to cover the distance between two consecutive areas $a, a+1 \in A_n$;
- t_n^{OUT} , is the time needed for a truck belonging to tuple $n \in N$ to cover the distance from the last area (i.e., the area where to perform its last operation) to the gate for exiting the terminal;
- Q_s^a , is the maximum number of moves available in area $a \in A$ during time slot $s = 1, \dots, S$. This depends on the availability of yard resources once the vessels requirements have been fulfilled, since the vessel cycle is prior for a container terminal.

The decision variables of the problem are represented by:

- $x_{ns} \in \{0, 1\}$, is a binary variable taking on value 1 if tuple $n \in N$ is assigned to time slot $s = 1, \dots, S$, 0 otherwise;
- $\delta_n \geq 0$, is a continuous variable representing the deviation between the preferred time slot of a truck associated with tuple $n \in N$ and the corresponding time slot assigned by the TAS;

- $L_{ns} \geq 0$, is a continuous variable to compute the truck turnaround time of tuple $n \in N$ when assigned to time slot $s = 1, \dots, S$.

The problem statement follows.

Problem

Find the optimal values of variables x_{ns} , δ_n and L_{ns} in order to first minimize the total deviation between trucks preferred time slots and their assigned slots and, later, to minimize truck turnaround times.

$$\text{Lex1 Min } \left\{ \sum_{n \in N} \alpha_n \delta_n, \sum_{n \in N} \sum_{s=1}^S L_{ns} \right\} \quad (2)$$

subject to:

$$\sum_{s=1}^S x_{ns} = 1 \quad \forall n \in N \quad (3)$$

$$\sum_{s=1}^S I_s \cdot x_{ns} - g_n \leq \delta_n \quad \forall n \in N \quad (4)$$

$$\sum_{s=1}^S F_s \cdot x_{ns} - g_n > -\delta_n \quad \forall n \in N \quad (5)$$

$$\delta_n \leq \delta_{\max} \quad \forall n \in N \quad (6)$$

$$\left(t_n^{IN} + \sum_{a \in A_n} \left(\frac{u_n^a}{\rho_s^a \beta_s^a} 60 + t_n^{a,a+1} \right) + t_n^{OUT} \right) \cdot x_{ns} \leq L_{ns} \quad \forall n \in N, s = 1, \dots, S \quad (7)$$

$$\sum_{n \in N: a \in A_n} u_n^a x_{ns} \leq Q_s^a \quad \forall a \in A, s = 1, \dots, S \quad (8)$$

$$x_{ns} \in (0, 1) \quad \forall n \in N, s = 1, \dots, S \quad (9)$$

$$\delta_n \geq 0 \quad \forall n \in N \quad (10)$$

$$L_{ns} \geq 0 \quad \forall n \in N, s = 1, \dots, S \quad (11)$$

The first objective function (2) of this lexicographical optimization model aims to minimize the total deviation of tuples/trucks from their preferred time slots. The parameter α_n allows to give preference to trucks performing double moves. Once this first goal is achieved, and if there are alternative optima, the second objective function aims to minimize truck turnaround times. Constraints (3) make sure that each tuple is served, whereas constraints (4) and (5) compute, in the first objective function, the deviation in delay and advance, respectively, when tuple n is assigned to time slot s . Constraints (6) bound the value of δ_n for each tuple n considering the maximum acceptable deviation $\delta_{\max}$ from truck preferred time slots. The total truck turnaround time of truck/tuple n when assigned to time slot s is computed by constraints (7). Here, $\frac{u_n^a}{\rho_s^a \beta_s^a} 60$ is the time needed for a truck belonging to tuple $n \in N$ to be served in area $a \in A_n$. Constraints (8) set a capacity condition establishing that in area a at most Q_s^a moves may be performed during time slot s . Finally, constraints (9), (10), and (11) define the nature of the problem decision variables.

The lexicographical method is a well-known technique of multiobjective optimization. With the lexicographic method, preferences are imposed by ordering the objective functions according to their importance or significance, rather than by assigning weights. In this way, the optimization problems are solved one at a time for each objective function, considering a predefined order (Arora, 2004). In our case the lexicographical model (2)–(11) is solved as follows:

- step 1: Minimize $\sum_{n \in N} \alpha_n \cdot \delta_n$, subject to Eqs. (3)–(22), where x_{ns} , δ_n and L_{ns} are decision variables.
- step 2: Minimize $\sum_{n \in N} \sum_{s=1}^S L_{ns}$, subject to Eqs. (3)–(9), (22), where x_{ns} and L_{ns} are decision variables, while δ_n is a parameter whose value is obtained from step 1.

It is important to underline that the proposed approach is valid both for container terminals provided and not provided with a TAS; in the former case, the truck preferred time slot is used to determine the deviation from the assigned time slot, while in the latter the gate-in time of the truck can be used alternatively. In case the terminal is not provided with a TAS, the proposed approach can help terminal managers in evaluating the benefits they would obtain if implementing one.

4. Experimental design analysis

All experimental results have been generated and analyzed using a laptop Intel Core i7 2.4 GHz with 16 GB of RAM. We have implemented the model in OPL programming language (version 12.6.1) and used CPLEX commercial solver (version 12.6.1) to solve it. Moreover, we have used MINITAB to carry out the statistical analysis.

4.1. Factorial design

In this section a full randomized factorial design study 2^4 is proposed with the aim of validating the proposed approach and appraising the significance/impact of four key factors on three performance metrics in case of implementing a TAS. Data mining and statistical techniques have been used to understand the system behavior in terms of subsystem interactions and factors influencing the problem objectives.

The following performance metrics have been considered:

Metric M1: the total delay of the appointments respect to their preferred time slot, computed as:

$$\sum_{n \in N} \delta_n$$

Metric M2: the average resource utilization (expressed in percentage %), computed as:

$$\sum_{a \in A} \sum_{n \in N} \sum_{s \in S} \frac{u_n^a x_{ns}}{Q_s^a} * 100$$

Metric M1: the average truck turnaround time, computed as:

$$\sum_{n \in N} \sum_{s \in S} \frac{L_{ns}}{N}$$

In addition, four factors have been studied:

- F1– *Truck arrival patterns* {LOW- uniform with low peaks, HIGH- high peaks}, which regards the patterns of trucks arriving at the terminal.
- F2– *Double moves quota* {LOW- 20%, HIGH- 95%}, that represents the portion of trucks that performs double moves.
- F3– *Resource availability* {LOW- 20% over the total demand of terminal moves, HIGH- 50% over the total demand of terminal moves}, that represents the container terminal resources available for the truck cycle (we have already pointed out that in a container terminal the priority is given to the vessel cycle).
- F4– *Yard organization* {LOW- u_n^a disperse, HIGH- u_n^a compact}, which represents the quality/goodness of the stacking process in terms of container allocation to terminal areas.

Usually, truck arrival patterns (factor F1) in real container terminals are characterized by the presence of some peaks during the gate opening time windows, which means that most of the trucks prefer to arrive in certain time periods. Unfortunately, the presence of peaks significantly affect the terminal performance. The use of a TAS can substantially help terminals in flatten their truck arrival distributions (i.e. obtain uniform arrival pattern) so to allow it to better exploit its resources. Note that factor F1 is associated with the g_n parameter, which represents the preferred time slot of tuple $n \in N$.

As already stated, one of the objective of our approach is to reduce the number of empty trips; this is obtained when a truck carrying 1 or 2 export containers, returns with 1 or 2 import containers. To do that, in the objective function, we weigh more the trucks performing double cycles. If tuple n is involved in a single cycle (i.e., only import or only export), its α_n value will be equal to 1, whereas if the tuple performs a the double cycle, so avoiding empty trips, α_n will assume the value of 2. However, the achievement of this objective, i.e., reducing empty trips, largely depend on the percentage of shippers/freight forwarders agreeing to share trucks so to perform double cycles. Therefore, Factor F2 (i.e., double moves quota) is associated with the attribute “agreement” (v). To make it clearer, let us consider two cases: in Case A, 73% of the containers have an agreement (i.e., they agree to be part of the same trip/tuple) while, in Case B, only 40% of the containers are available to collaborate. This means that, in case A, at least 27% of containers are performed in single cycle, so generating empty trips. This percentage raises up to 60% in Case B. In other words, if the other criteria used to form tuples allow it, in Case A there will be a larger number of tuples performing double moves than in Case B. In [Appendix C](#), a more precise example is provided to further clarify this point.

[Table 3](#) shows how the average deviation from the preferred time slot δ_n varies when using different values of α_n .

More specifically, [Table 3](#) shows, for each case, the portion of trucks deviated from their preferred time slots, when varying the capacity (availability of terminal resources), α_n (priorities) and maximum allowed deviations (δ_{max}). [Fig. 4](#) summarizes the results of [Table 3](#) for single (Cases 1 to 4) and double (Cases 5 to 8) moves. Three alternatives have been assessed regarding the values of α_n : (1) the *Current alternative* differentiates trucks performing single and double moves (i.e., $\alpha_n = 1$ for Cases 1 to 4, and $\alpha_n = 2$ for Cases 5 to Case 8), (2) *Alternative 1* presents more differences between cases (i.e., $\alpha_n = 1$ for Cases 1 and 3; $\alpha_n = 2$ for Cases 2 and 4; $\alpha_n = 3$ for Cases 5, 6 and 7, and $\alpha_n = 4$ for Case 8), while (3) *Alternative 2* does not consider differences between cases (i.e., $\alpha_n = 1$ for all cases). The obtained results highlight that, for Alternative 2, double moves exhibit greater deviations; this could be due to the fact that, as double moves require more terminal resources, and having no priority over single moves, they are programmed with a greater deviation from their preferred time slots.

For the remaining alternatives (*Current* and *Alternative 1*), when δ_{max} increases, the portion of trucks deviated from their preferred time slots increases, since trucks are more willing to change their preferred schedule. On the contrary, an extreme low value of δ_{max} (near to zero) results in infeasible solutions. As expected, the availability of terminal resources affects the deviation from truck preferred time slots in a different way between trucks performing single and double moves. In all cases, the portion of trucks deviated from their preferred time slots is lower when the terminal capacity is high. It can be also noted that, when a truck performs a double move, its deviation from its preferred time slot is lower than when it performs a single move; this is due to the fact that for alternative *Current* and *Alternative 1* we give priority to double moves.

Factor F3 is related to the container terminal resource availability. This factor is associated with Q_s^a parameter (i.e., the maximum

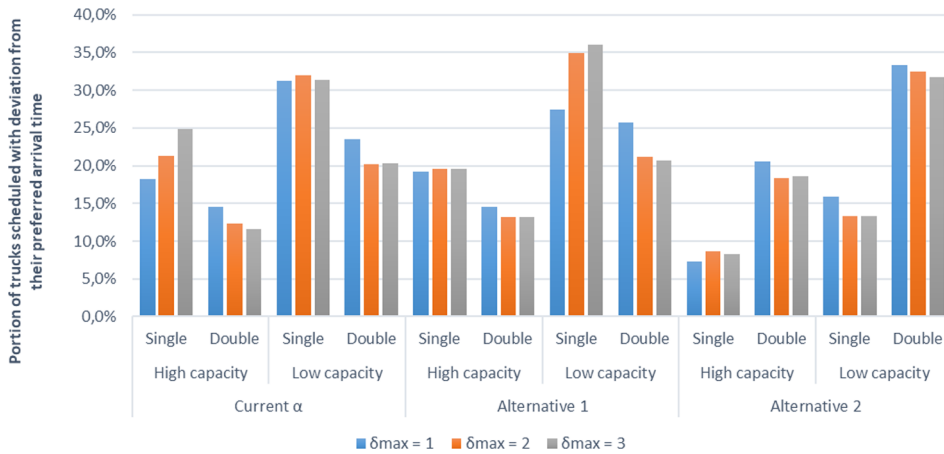


Fig. 4. Sensitivity analysis considering different terminal capacity (high-low), priorities (α_n) and maximum allowed deviations (δ_{max}).

number of moves available in each area and time slot), and depends on the total number of moves required by the set of tuples (u_n^a). It should be noted that $V_s = \sum_{a \in A} Q_s^a$ represents the appointment quota for time slot s . This value may vary from a time slot to another since it may happen that, during the loading/unloading of a ship (or a train), the terminal may reduce the maximum number of appointments offered.

Factor F4 is related to the goodness of the yard organization in terms of container allocation to terminal areas; this factor is also associated with parameter u_n^a , and it is an indicator of how efficient was the clustering phase to form the tuples (efficiency of the clustering approach). A good clustering will result in few areas to be visited by a truck, also close to each other.

Finally, both factors F1 and F2 reflect the degree of cooperation between the stakeholders involved in the port cycle (truckers, freight forwarders, shippers, receivers, container terminal).

Table 4 shows the features of the 16 scenarios considered (in fact, it is a 2^4 factorial design study), where the symbols – and + indicate respectively low and high level for the corresponding factor. Each scenario has been replicated 25 times, which makes a total of 400 problem instances. To analyze the significance of the four factors on the three metrics, in this experimental part we did not bind the value of δ_{max} , setting it to infinite, so not imposing any bound on any metric. Under this circumstance, the maximum deviation observed was $\delta_{max} = 6$. These instances were solved by the lexicographical model and the response variables were recorded in order to perform a statistical analysis.

An instance generator was coded in C programming language to manage the instances generation and analysis. All problem instances are available under request. All instances consider 300 tuples, which represent the movement of 300 (only one container managed in a trip) to 1200 (four containers managed in a trip) containers per day, picked-up and delivered in 10 terminal areas along 10 time slots of 1 h each. The computing time per instance ranges from 5.15 to 11.96 min (averaging 6.94 min), showing a good performance.

Let us first consider scenario 8, that represents the best situation: the arrival pattern of trucks is uniformly distributed along the gate opening hours (F1), 95% of trucks perform double moves (F2), 50% of terminal resources are dedicated to the truck cycle (F3), and containers are well distributed in the yard (F4).

Table 4
The considered scenarios.

Scenario	F1	F2	F3	F4
1	–	–	–	–
2	–	–	–	+
3	–	–	+	–
4	–	–	+	+
5	–	+	–	–
6	–	+	–	+
7	–	+	+	–
8	–	+	+	+
9	+	–	–	–
10	+	–	–	+
11	+	–	+	–
12	–	–	+	+
13	+	+	–	–
14	+	+	–	+
15	+	+	+	–
16	+	+	+	+

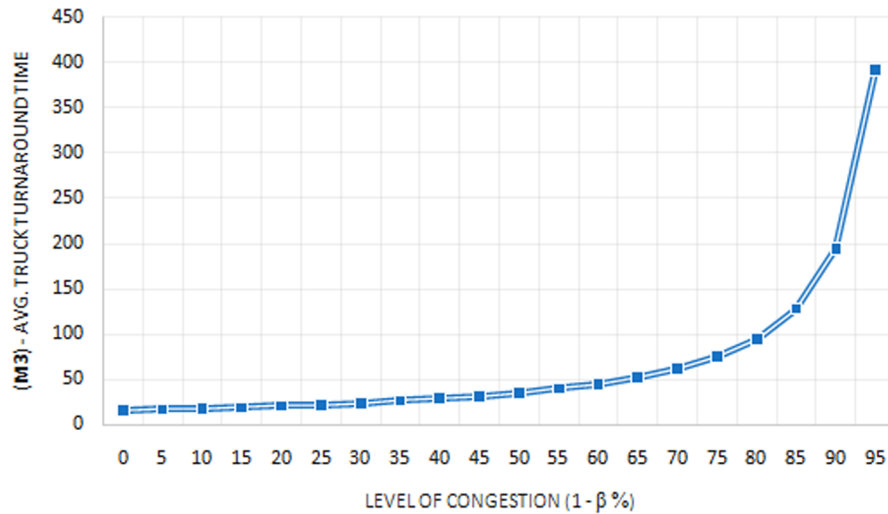


Fig. 5. The impact of congestion level on the Average Truck Turnaround Time (Avg. TTA).

Fig. 5 shows, for scenario 8, the impact of the degree of congestion in terminal container areas over the truck turnaround time (metric M3). More specifically, the degree of congestion in container areas ($1 - \beta_s^a$) has been varied from 0 (non congested) to 95% (highly congested): the results clearly point out that when the degree of congestion reaches values up to 60%, the truck turnaround time starts increasing exponentially.

The impact of the percentage of double appointments (factor F2) on the percentage of empty trips and of “shifted” trucks (i.e., both delayed or anticipated respect to their preferred time slot) is instead provided in Fig. 6, again for scenario 8. The obtained results indicate that when increasing factor F2, the quantity of empty trips decreases. In fact, on equal number of containers managed, a bigger number of double moves reflects a shorter quantity of trucks to perform the necessary operations at the container terminal. Moreover, when increasing the number of double appointments, the percentage of trucks delayed or anticipated respect to their preferred time slot decreases since there is a smaller number of trucks to be managed and, hence, the possibility of satisfying truckers’ schedule preferences increases.

After this preliminary analysis of the level of congestion and the number of double moves, in the following the behavior of each individual factor and their interactions will be analyzed, through the use of MANOVA (Multivariate Analysis of Variance) and ANOVA (Analysis of Variance) techniques.

Since the total delay (M1) and the average resource utilization (M2) are lightly correlated (their partial correlation is 0.4734), the statistical analysis was based on a MANOVA. This correlation means that when the average resource utilization increases (or the capacity of the terminal decreases per time slot), the deviation between the preferred time slot and the time slot assigned also increases. The MANOVA is used to determine whether there are any differences between independent groups (which in our case represent the scenarios) on more than one response variable (i.e., M1 and M2). Furthermore, the use of MANOVA minimizes the probability of making one or more types I errors for the entire set of comparisons.

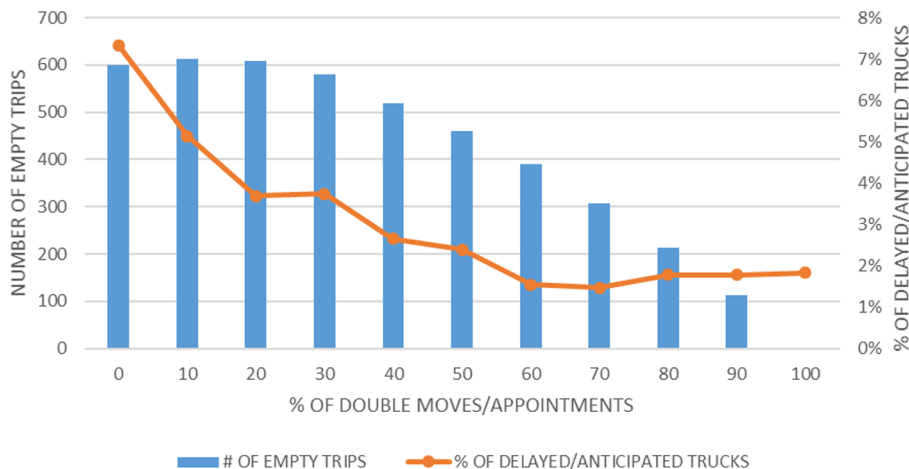


Fig. 6. The impact of double moves on the Average Truck Turnaround Time.

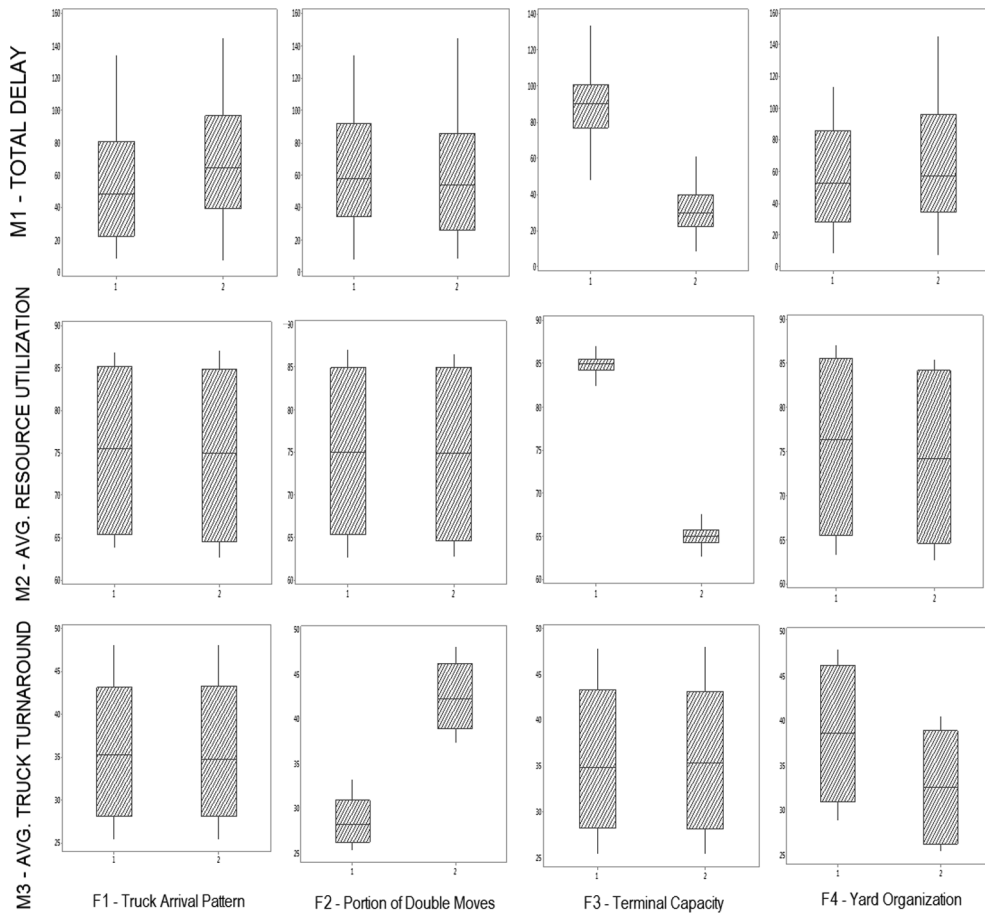


Fig. 7. Boxplots: impacts of Factors F1-F4 over metrics M1-M3.

The boxplot depicted in Fig. 7 shows that the total delay (metric M1) and the average resource utilization (metric M2) are mainly affected by the container terminal availability of resources (factor F3), while the average truck turnaround time (metric M3) is primarily influenced by the portion of double moves (factor F2) and the efficiency of the clustering approach (factor F4). In the case of compact distribution of containers in double moves (factor F4 = 2), the average truck turnaround time (Avg.TTA) is notably smaller than the truck turnaround time in disperse scenarios. In fact, if import and export container are well stored within the yard, smaller truck turnaround time will be expected.

The *Main Effects* plots depicted in Fig. 8 show the impact of each factor on the response variables (i.e., metrics M1, M2 and M3); as expected, we obtained the same insights provided by Fig. 7. Moreover this graph, in addition to the magnitude of the impact of each factor, also highlights their directions of impact. For instance, Factor F1 affects metric M1 in a positive way, while Factor F3 affects the same metric in a negative way. Factors F2 and F4 seem not to have any impact on metric M1, but the confirmation will be given only after performing ANOVA, as shown later in the paper.

To complement the MANOVA analysis, an ANOVA analysis was performed for each individual response variable. More specifically, each analysis of variance evaluates the effects of the four factors and their two levels interactions on each response variable. When using ANOVA, three main assumptions must be checked: normality, homogeneity of variance, and independence of residuals. We did these checks and found no basis for questioning the validity of the experiments.

Table 5 shows the results of the ANOVA, i.e., the effects of the studied factors on the total delay (M1). The obtained results ($R^2 = 85.67\%$) show that all factors and the two-way interaction of F1 (truck arrival pattern) and F2 (portion of double moves) are statistical significant. The terminal resource availability (F3) is the main contributor for the total delay (M1), since it explains 77.56% of the variance. The interaction between the truck arrival pattern and the portion of double moves (F1*F2) is also significant; therefore, we cannot consider the effects of both factors separately.

The ANOVA in Table 6 shows the effects of the four factors on the average resource utilization (M2). These ANOVA results ($R^2 = 99.61\%$) show that the container terminal capacity (F3) and the organization of the yard (F4) are the main responsible of this performance metric. In other words, a significant amount of variation (99.40%) in the average resource utilization is associated with a variation in both factors.

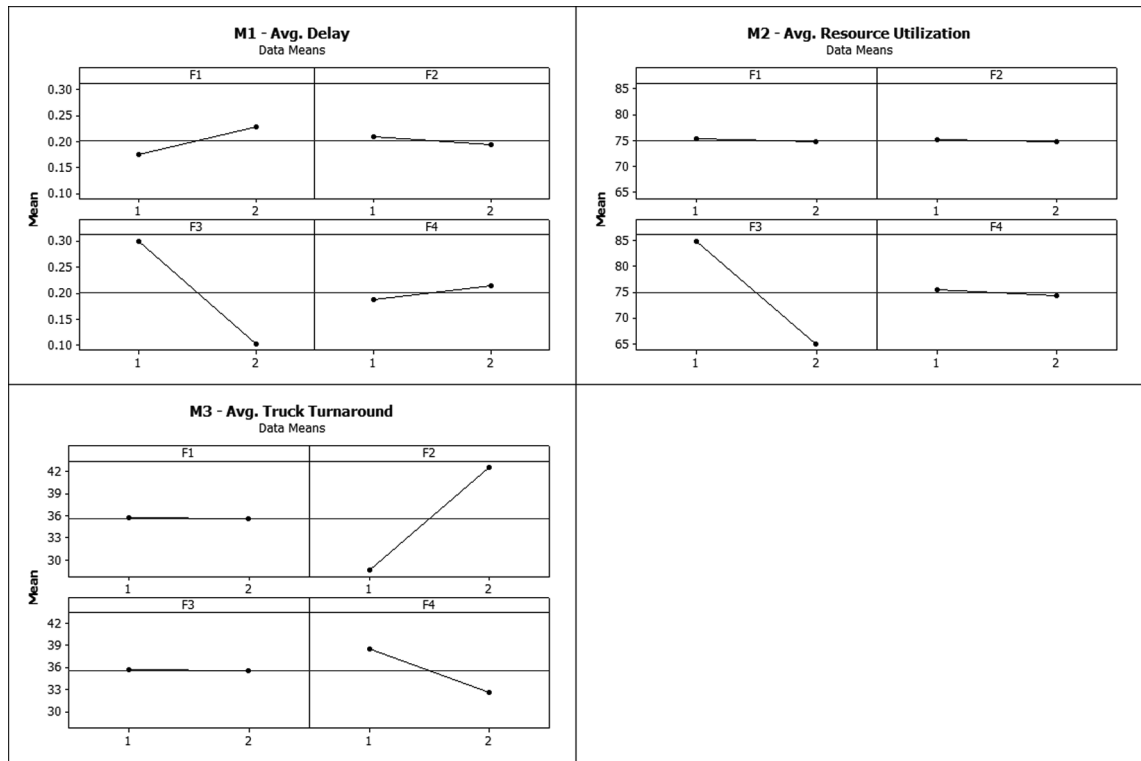


Fig. 8. Main Effects Plots for metrics (response variables) M1, M2 and M3.

Table 5
ANOVA for total delay.

Source	df	sum of squares	Mean Square	p-value	Contri. %	Cum. Contri. %
F3	1	345803	345803	0.0000	77.56	77.56
F1	1	23731	23731	0.0000	5.32	82.88
F4	1	5845	5845	0.0000	1.31	84.19
F1*F2	1	2637	2637	0.0000	0.59	84.78
F2	1	1862	1862	0.0010	0.42	85.20
F3*F4	1	1395	1395	0.0040	0.31	85.52
F2*F4	1	536	536	0.0720	0.12	85.64
F2*F3	1	120	120	0.3930	0.03	85.66
F1*F3	1	21	21	0.7230	0.00	85.67
F1*F4	1	17	17	0.7460	0.00	85.67
Error	389	63889	164		14.33	100.00
Total	399	445855				

R-squared = 0.8567 (adjusted R-squared = 0.8530).

The ANOVA in Table 7 shows the effects of the factors on the average truck turnaround (M3). The results ($R^2 = 99.08\%$) point out that the portion of double moves (F2), the organization of the yard (F4), and the interaction between them mainly affect this performance metric, explaining the 99.07% of the observed variability.

Once performed the ANOVA analysis for each response variable independently, the MANOVA analysis has been executed. To assess the performance of the MANOVA, the variability, error, and partial correlations matrices have been examined. The correlation structure between the response variables was robust enough, indicating the adequacy of the MANOVA. To appraise how the response variables are related, a partial correlation between them has been computed. Table 8 shows that the total delay (M1) and the average resource utilization (M2) are weakly correlated (partial correlation equal to 0.4356).

To judge where there is significant evidence for the model effects, the Wilks' test has been used. In Table 8, the p-values show that almost all factors and two-way interactions affect the responses differently. The relative contribution of each factor over each response variable is shown. The truck arrival pattern (F1) is the least significant factor in every response variable.

The Eigen analysis was then used to assess how the response means differ among the levels of the different factors. From the results of the Eigen analysis, it is evident that metric M2 (average resource utilization) presents the largest difference among the factor levels for all factors and interactions. The total delay (M1) has the smallest differences for all the factors.

Table 6
ANOVA for average resource utilization.

Source	df	sum of squares	Mean Square	p-value	Contri. %	Cum. Contri. %
F3	1	39364.8	39364.8	0.0000	99.06	99.06
F4	1	134.3	134.3	0.0000	0.34	99.40
F1	1	26.5	26.5	0.0000	0.07	99.46
F2*F4	1	21.9	21.9	0.0000	0.06	99.52
F2*F3	1	12.8	12.8	0.0000	0.03	99.55
F2	1	8.3	8.3	0.0000	0.02	99.57
F1*F3	1	7.7	7.7	0.0000	0.02	99.59
F3*F4	1	7.0	7.0	0.0000	0.02	99.61
F1*F2	1	0.0	0.0	0.7450	0.00	99.61
F1*F4	1	0.0	0.0	0.8070	0.00	99.61
Error	389	155.3	0.4		0.39	100.00
Total	399	39738.7				

R-squared = 0.9961 (adjusted R-squared = 0.9960).

Table 7
ANOVA for average truck turnaround time.

Source	df	sum of squares	Mean Square	p-value	Contri. %	Cum. Contri. %
F2	1	19552.9	19552.9	0.0000	83.47	83.47
F4	1	3476.2	3476.2	0.0000	14.84	98.31
F2*F4	1	177.0	177.0	0.0000	0.76	99.07
F1	1	0.9	0.9	0.1920	0.00	99.07
F3*F4	1	0.9	0.9	0.2120	0.00	99.08
F2*F3	1	0.7	0.7	0.2550	0.00	99.08
F3	1	0.5	0.5	0.3520	0.00	99.08
F1*F3	1	0.1	0.1	0.6050	0.00	99.08
F1*F2	1	0.1	0.1	0.6790	0.00	99.08
F1*F4	1	0.0	0.0	0.9210	0.00	99.08
Error	389	215.1	0.6		0.92	100.00
Total	399	23424.5				

R-squared = 0.9908 (adjusted R-squared = 0.9906).

Table 8
MANOVA results.

Source	Wilks' Test p-value	Variance contribution (%)			Eigenvectors		
		M1	M2	M3	M1	M2	M3
F1	0.000	5.32	0.07	0.00	0.0035	−0.0564	0.0009
F2	0.000	0.42	0.02	83.47	0.0004	−0.0014	0.0687
F3	0.000	77.56	99.06	0.00	0.0003	−0.0816	−0.0017
F4	0.000	1.31	0.34	14.84	0.0000	0.0206	0.0664
F1*F2	0.001	0.59	0.00	0.00	0.0040	−0.0255	0.0017
F1*F3	0.000	0.00	0.02	0.00	0.0006	−0.0818	0.0071
F2*F3	0.000	0.03	0.03	0.00	0.0002	−0.0796	−0.0151
F2*F4	0.000	0.12	0.06	0.76	0.0004	−0.0305	0.0634
F3*F4	0.000	0.31	0.02	0.00	0.0027	−0.0705	−0.0104
Error		14.33	0.39	0.92			

5. Case studies: Italian and Mexican container terminals

In order to further assess the performance of the proposed approach, two different real case studies have been considered and analyzed: the container terminal of PSA – Genova Pra located in the Italian port of Genoa and the container terminal APT in the Mexican port of Altamira. Both terminals provided us with datasets containing the records of the containers picked-up and delivered during a specific time period (59,570 records related to the last four months of 2017 for the Mexican case, and 114,399 records corresponding to the last two months of 2017 for the Italian case). These records were processed following the Knowledge Discovery in Databases process (Fayyad et al., 1996) (i.e., data selection, preprocessing, transformation, data mining, evaluation and knowledge extraction), looking for inconsistencies, missing values and outliers. Additionally, for each case study, we reviewed the day by day behavior of the truck arrival patterns over the corresponding period, looking for an instance that resembles the behavior of the truck arrival pattern and average truck turnaround time in each port reality in a typical day. For the Mexican case, we selected an instance of 415 containers (314 export and 101 import) that were moved by trucks in a significant day, while for the Italian case, we



Fig. 9. A picture of ATP Terminal.

considered an instance of 2,466 containers (1,191 export and 1,275 import) that were managed in a significant day.

Regarding the clustering phase, attributes (i) cycle, (ii) ISOtype, (iii) weight, (iv) special, (v) vessel departure time, (vii) distance, (viii) customs clearance and (ix) dwell timewere provided by the two container terminals. As far as concerns attribute (v) agreement, we determined the portion of containers with agreement based on the data set and the interviews with terminal operators, so in a semi-random way. Attribute (x) destination has been semi-randomly determined since container terminals do not have this information but Port Authorities provide the main hinterland areas of the related ports.

Moreover, the dataset provided by each container terminal includes the following information: type of cycle, check-in time, gate-in time, gate-out time, time at which each operation (pick up or deliver) is executed in the terminal yard, truck ID, container ID, type of container, ID of the location where the container was picked-up or delivered. Additionally, some meetings with the terminal container managers were executed in order to gather information regarding: (i) the average productivity of yard resources, (ii) the levels of congestion in the yard areas, (iii) the maximum number of moves available in each area per time slot (both related with planned work and vessels activity), and (iv) the layouts and road networks inside the terminal in order to compute the travel time matrices. With all this information at hand we created the instances for these two case studies.

5.1. Mexican Case: Altamira port terminal

The port of Altamira in Mexico is located in the Gulf of Mexico; it is the second largest Mexican port on the Gulf of Mexico moving, on average, more than 600,000 containers per year and connected to over 20 major ports worldwide. Altamira Port Terminal (ATP, its acronym is in Spanish) is one of the two container terminals operating in the port of Altamira. Approximately 60% of the cargo passing through ATP is transported within a radius of about 120 km from the port, and 40% within a radius of 500 km.

A picture of ATP terminal is provided in Fig. 9. This terminal manages around 324 trucks per day, on average.

Fig. 10 shows the behavior of the truck arrival pattern and the average truck turnaround time of ATP terminal for nine weeks of 2017. Fig. 11 provides, instead, the behavior of the same variables for the particular selected instance.

Clustering results.

As shown in Table 9, to pick up or deliver the 415 containers of the chosen instance, 299 trucks were required in the real case. By analyzing them, it comes out that 111 (37.12%) of the arriving trucks carried two import or two export containers (cases 2, 4, 6, 7 and 8), but only 4 (1.34%) of them performed a double move/appointment (case 5 to 8); 295 (98.66%) trucks generated an empty trip (given by the sum of cases 1, 2, 3 and 4). The average truck turnaround time was 21.53 min, but when the truck performed a double cycle it reached an average of 42.15 min.

After applying the clustering phase (with a threshold value $\theta = 0.85$), it came out that, to move the same amount of containers (i.e., 415), only 218 trucks would be needed instead of 299 and the number of empty trips would be reduced from 295 to 149, as depicted in Table 9.

Optimization results.

After performing the clustering analysis on the Mexican selected instance, the optimization model was applied considering the 218 tuples. Each tuple identifies the number of import and export container to move, accordingly to their case. Some parameters of the optimization model were considered as not fixed and several instances of the problem based on such parameters were generated; these parameters are the number and location of the areas to be visited (the u_n^a parameter), the preferred time slot (g_n) of each tuple (that was randomly generated according to the current truck arrival pattern), the sequence of areas to be visited by a tuple (A_n). The remainder parameters, i.e., yard crane productivities, expected congestion levels, distance between areas, and travel times between

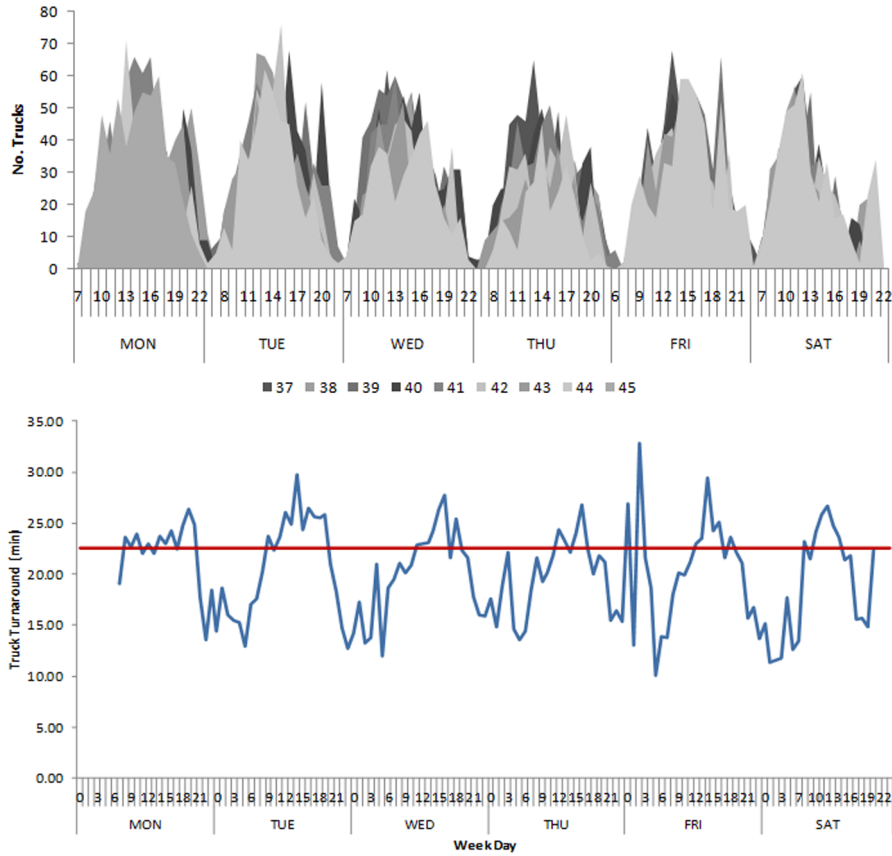


Fig. 10. Real distribution of trucks along gate opening window at ATP container terminal over 9 weeks of 2017.

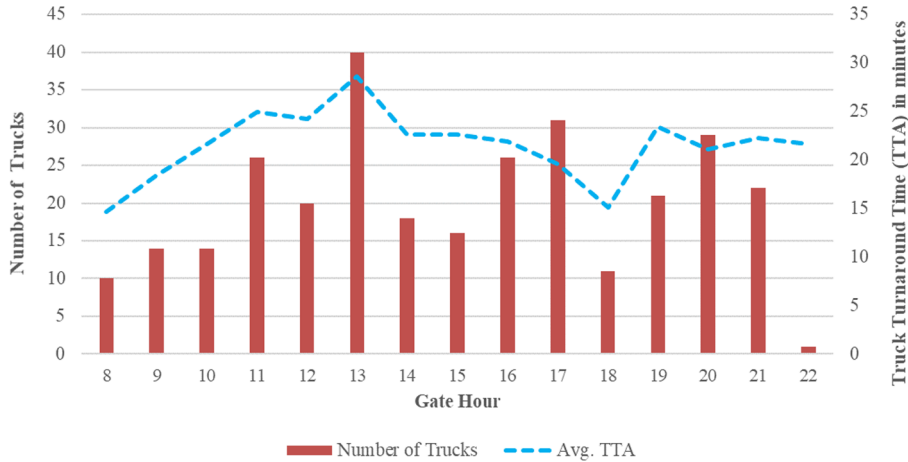


Fig. 11. Mexican instance: distribution of trucks along gate opening window at ATP container terminal in a typical day.

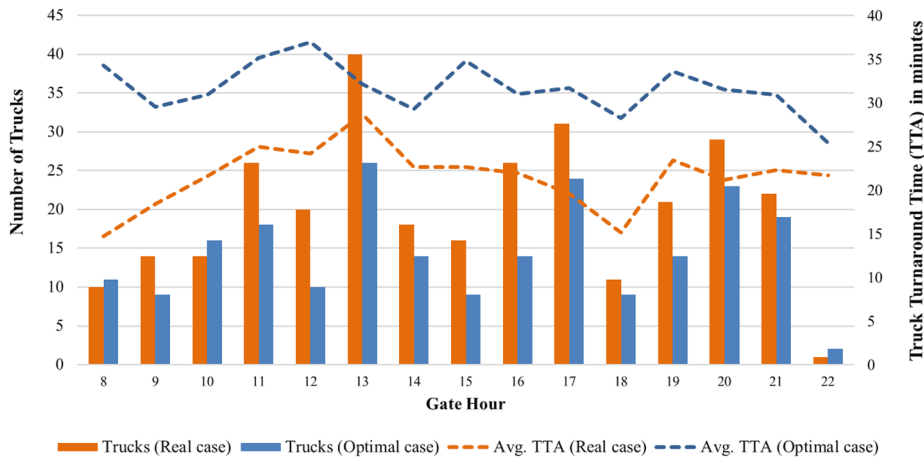
yard and gates, were obtained from the records and interviews with the container terminal staff and they were instead associated with fixed values. Regarding the maximum deviation allowed from the truck preferred time slot, a big Mexican transportation company working with the container terminal of Altamira suggested us to set $\delta_{max} = 4$.

The analysis of the results obtained leads to the major conclusion that the proposed approach outperforms the one currently in use. In fact, Fig. 12 shows that the number of trucks needed and, consequently, the number of empty trips, substantially improves, decreasing from 295 (98.66%) to 149 (68.35%). The number of trucks required to move the 415 containers of the selected instance decreases by 27.09%. Although the average truck turnaround time increases up to 47.11% (from 21.53 to 31.67 min) due to the increasing of double moves operations, its variance decreases by 30.34%.

Table 9

Comparison between real data and clustering analysis results for the Mexican instance.

Cluster Type	(Exp, Imp)	Export Containers		Import Containers		Trucks	
		Real Data	Clustering Analysis	Real Data	Clustering Analysis	Real Data	Clustering Analysis
Case 1	(1, 0)	127	67	—	—	127	67
Case 2	(2, 0)	182	148	—	—	91	74
Case 3	(0, 1)	—	—	59	8	59	8
Case 4	(0, 2)	—	—	36	0	18	0
Case 5	(1, 1)	2	34	2	34	2	34
Case 6	(1, 2)	1	5	2	10	1	5
Case 7	(2, 1)	0	22	0	11	0	11
Case 8	(2, 2)	2	38	2	38	1	19
TOTAL		314	314	101	101	299	218

**Fig. 12.** Results obtained by applying the proposed approach to a selected instance of ATP Mexican container terminal.

5.2. Italian Case: PSA – Genova Pra Italian container terminal.

The container terminal of PSA – Genova Pra (ex Voltri Terminal Europa-VTE) is located in the Ligurian port of Genoa, in Italy. Since 1998 the terminal is part of PSA International, one of the world leader companies in the field of container port terminal management. With a throughput of 1.6 million TEUs in 2018, it represents one of the largest container terminals in the Mediterranean basin.

A picture of PSA – Genoa Pra terminal is provided in Fig. 13. The terminal manages around 2500 trucks per day, on average.

Fig. 14 shows the behavior of the truck arrival pattern and the average truck turnaround time of the terminal for the last two months in 2017, whereas Fig. 15 shows the same variables for the chosen instance.

Clustering results.

The analysis of the real data showed that, in order to move 2,466 containers in the chosen day, 1,961 trucks were required, as shown in Table 10. A deeper analysis revealed that 67 (3.42%) arriving trucks carried two import or export containers (cases 2, 4, 6, 7 and 8), and 436 (22.23%) performed double moves (case from 5 to 8); moreover, 1,525 (77.77%) trucks generated an empty trip in the real scenario. The average truck turnaround time is 39.36 min, that raises to 52.87 min when a double cycle is performed.

After applying the clustering phase (with a threshold value $\theta = 0.87$), it results that a lower number of trucks (1,155 against 1,961) is necessary to move the 2,466 required containers, as described in Table 10.

From Table 10 it can be seen that the clustering analysis increases both the number of trucks carrying two import or export containers (from 67 to 278 (75.90%)) and the number of trucks performing double moves (from 436 to 910 (52.09%) trucks).

Optimization results.

As for the Mexican case, the results obtained by applying the optimization model to the instance of PSA – Genoa Pra terminal highlight that the proposed approach outperforms the one currently in use. In this case, an important transportation company working with the container terminal of Genova suggested us to set $\delta_{max} = 3$. As a matter of fact, as shown in Fig. 16, the number of empty trips decreases from 1525 (77.77%) to 245 (21.21%) and the number of trucks required to move the 2466 containers reduces by 41.10%. Although the average truck turnaround time increases up to 8.23% (from 39.36 to 42.60 min), its variance decreases by 98.94%.



Fig. 13. A picture of PSA – Genova Pra container terminal.

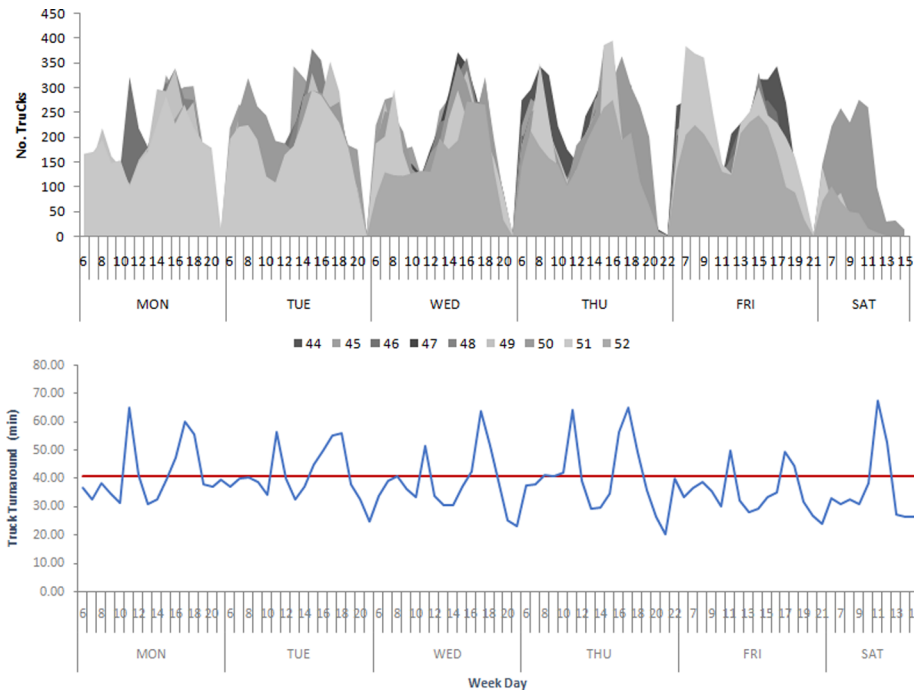


Fig. 14. Real distribution of trucks at PSA container terminal over the last two months of 2017.

Finally it must be observed that, in both cases (Mexican and Italian terminals), the truck turnaround time is lower than the one associated with the double moves in the real cases.

6. Conclusions

In this paper we propose a novel methodological framework to reduce empty truck trips and improve truck service levels (in terms of minimization of the deviation from their preferred time slots and turnaround times) in container terminals provided with a truck appointment system. The provided approach combines data mining (DM) and operational research (OR) techniques, through a clustering analysis and a lexicographical optimization model.

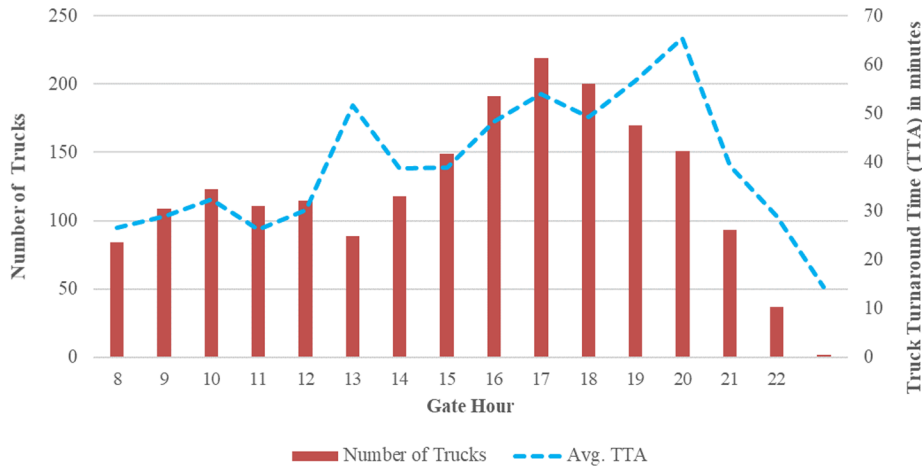


Fig. 15. Italian instance: distribution of trucks along gate opening window at PSA container terminal in a typical day.

Table 10

Comparison between real data and clustering analysis results for the Italian instance.

Cluster Type	(Exp, Imp)	Export Containers		Import Containers		Trucks	
		Real Data	Clustering Analysis	Real Data	Clustering Analysis	Real Data	Clustering Analysis
Case 1	(1, 0)	720	127	—	—	720	127
Case 2	(2, 0)	30	4	—	—	15	2
Case 3	(0, 1)	—	—	749	112	749	112
Case 4	(0, 2)	—	—	82	8	41	4
Case 5	(1, 1)	425	638	425	638	425	638
Case 6	(1, 2)	6	122	12	244	6	122
Case 7	(2, 1)	6	54	3	27	3	27
Case 8	(2, 2)	4	246	4	246	2	123
TOTAL		1191	1191	1275	1275	1961	1155

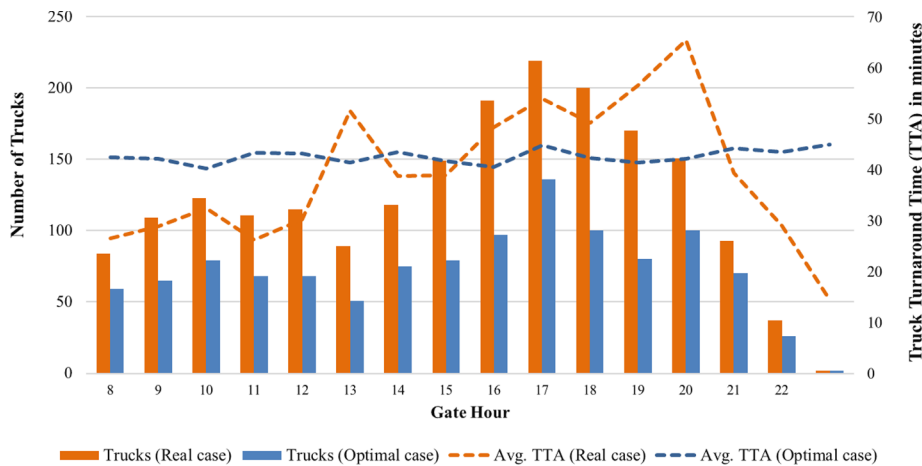


Fig. 16. Results obtained by applying the proposed approach to a selected instance of PSA-Genova Pra Italian container terminal.

An extensive experimental campaign, including a factorial design analysis and a sensitivity analysis on specific parameters, has been carried out to test the goodness of the proposed approach and to better understand how some factors influence the obtained solutions.

The results obtained pointed out that all factors and some of their corresponding two-way interactions are statistically significant for the three metrics (or response variables) and that the container terminal availability of resources represents the main responsible for the variance of the three response variables. Moreover, the organization of the yard is the second most responsible factor for the

observed variability of the three variables.

The approach has also been successfully applied to two real container terminals of different size, a medium-sized terminal located in the Mexican port of Altamira and a big-sized one belonging to the Italian port of Genoa. In both cases, the proposed methodology was able to improve the terminal performances both in terms of empty trips reduction and truck service level enhancement (lower deviation from truck preferred time slots and lower turnaround times).

The proposed approach is beneficial for many stakeholders involved in the port logistics chain: by coupling more containers in the same trip, i.e. increasing double moves, truckers can perform more orders per day (i.e., containers to deliver) with the same number of trucks. This, in turn, reflects in a higher revenue for them since they gain per container and not per trip. Then, there is an obvious benefit for terminals that can better plan their activities and better exploit their resources/equipment. Finally, a better planning of trips and the increase of double moves also allows to reduce times of delivery, and this constitutes an advantage also for receivers that can have their goods at disposal in shorter times.

Finally, it is worth underlining that the paper results represent an incentive for port stakeholders to cooperate and the approach is general enough to be applied to any container terminal located worldwide.

Credit authorship contribution statement

Claudia Caballini: Conceptualization, Methodology, Investigation, Writing - original draft, Visualization, Project administration.
Maria D. Gracia: Conceptualization, Methodology, Formal analysis, Writing - review & editing, Visualization.
Julio Mar-Ortiz: Conceptualization, Methodology, Software, Formal analysis, Writing - original draft, Project administration.
Simona Sacone: Conceptualization, Methodology, Investigation, Visualization, Writing - review & editing, Supervision.

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Appendix A. Another version of the mathematical model (Model 2)

This model aims to: (1) matches export containers two by two to form export tuples, when feasible; (2) matches import containers two by two to form import tuples, when feasible; (3) matches an export and an import tuple to form complete tuples; and (4) assigns tuples/trucks to time slots.

Due to the illustrative purpose of this alternative integrated optimization model, without loss of generality, we omit to formulate constraint (7), and only concentrate on the first objective of our lexicographical model.

For modeling purposes, we use the following notation:

- A , is the set of terminal areas;
- S , is the set of time slots;
- T , is the set of trucks available, $t = 1, \dots, T$;
- E , is the set of export containers, $i = 1, \dots, n_E$;
- I , is the set of import containers, $k = 1, \dots, n_I$;
- $a(i)$, is the area where export container i will be delivered;
- $a(k)$, is the area where import container k will be picked-up;
- $\rho(i) = 1$, if there is a collaboration agreement associated with import container i , 0 otherwise;
- $\rho(k) = 1$, if there is a collaboration agreement associated with export container k , 0 otherwise;
- $g(t)$, is the preferred time slot at which the truck t would like to arrive at the terminal;
- Q_s^a , is the maximum number of moves available in area a during time slot s ;
- $U(i)$ denote the set of containers that are adjacent to container i . A container j is adjacent to container i if both containers can be matched together. This parameter may be estimated manually or it can be computed using a similarity coefficient, note that $j \in U(i)$ if and only if $S_{ij} \geq \theta$.

The decision variables are represented by:

- $x_{ij}^E \in \{0, 1\}$, is a binary variable taking on value 1 if export containers i and j are transported together, 0 otherwise. Note that when $x_{ii}^E = 1$ export container i is transported alone.
- $x_{kl}^I \in \{0, 1\}$, is a binary variable taking on value 1 if import containers k and l are transported together, 0 otherwise. Note that when $x_{kk}^I = 1$ import container k is transported alone.
- $y_{ij}^E \in \{0, 1\}$, is a binary variable taking on value 1 if the truck transporting the export containers i and j do not perform a double move (returns in an empty trip), 0 otherwise. Note that when $y_{ij}^E = 1$ for $i = j$ it means that the truck only transports the export container i and then returns in an empty trip (Case 1), when $y_{ij}^E = 1$ for $i \neq j$ Case 2 results.
- $y_{kl}^I \in \{0, 1\}$, is a binary variable taking on value 1 if the truck transporting the import containers k and l do not perform a double

move (arrives in an empty trip), 0 otherwise. Note that when $y_{kl}^I = 1$ for $k = l$ it means that the truck only transports the import container k , so generating an empty trip (Case 3), when $y_{kl}^I = 1$ for $k \neq l$ Case 4 results.

- $y_{ij}^{kl} \in \{0, 1\}$, is a binary variable taking on value 1 if the truck transporting the export containers i and j perform a double move returning with import containers k and l , 0 otherwise. If $y_{ij}^{kl} = 1$ for $i = j$ and $k = l$ Case 5 results, if $y_{ij}^{kl} = 1$ for $i \neq j$ and $k = l$ Case 6 results, if $y_{ij}^{kl} = 1$ for $i = j$ and $k \neq l$ Case 7 results, finally, if $y_{ij}^{kl} = 1$ for $i \neq j$ and $k \neq l$ Case 8 results.
- $y_{ijst}^E, y_{klt}^I, y_{ijl}^{kl} \in \{0, 1\}$, are binary variables taking on value 1 if the corresponding tuple is assigned to time slot $s = 1, \dots, S$ in truck $t = 1, \dots, T$, 0 otherwise.
- $\delta_{ij}^E, \delta_{kl}^I, \delta_{ijl}^{kl} \geq 0$, are continuous variables representing the deviations observed, for export and import container respectively, between the preferred time slot and the corresponding assigned time slot.

The mathematical formulation follows.

Model 2:

$$\begin{aligned} \text{Min} \quad & \sum_{i=1}^{n_E-1} \sum_{j=i+1}^{n_E} x_{ij}^E + \sum_{k=1}^{n_I-1} \sum_{l=k+1}^{n_I} x_{kl}^I + \sum_{i=1}^{n_E-1} \sum_{j=i+1}^{n_E} \sum_{k=1}^{n_I-1} \sum_{l=k+1}^{n_I} y_{ij}^{kl} \\ & + \sum_{i=1}^{n_E-1} \sum_{j=i+1}^{n_E} \delta_{ij}^E + \sum_{k=1}^{n_I-1} \sum_{l=k+1}^{n_I} \delta_{kl}^I + \sum_{i=1}^{n_E-1} \sum_{j=i+1}^{n_E} \sum_{k=1}^{n_I-1} \sum_{l=k+1}^{n_I} 2 \cdot \delta_{ijl}^{kl} \end{aligned} \quad (12)$$

subject to:

$$\sum_{j \in U(i)} x_{ij}^E \leq 1 \quad \forall i = 1, \dots, n_E \quad (13)$$

$$\sum_{q \in U(i) \setminus q \neq j} x_{iq}^E + \sum_{p \in U(j) \setminus p \neq i} x_{pj}^E + x_{ij}^E \geq 1 \quad \forall i = 1, \dots, n_E - 1, j = i + 1, \dots, n_E \quad (14)$$

$$\sum_{l \in U(k)} x_{kl}^I \leq 1 \quad \forall k = 1, \dots, n_I \quad (15)$$

$$\sum_{q \in U(k) \setminus q \neq l} x_{kq}^I + \sum_{p \in U(l) \setminus p \neq k} x_{pl}^I + x_{kl}^I \geq 1 \quad \forall k = 1, \dots, n_I - 1, l = k + 1, \dots, n_I \quad (16)$$

$$\begin{aligned} y_{ij}^{kl} &\leq \frac{(x_{ij}^E + x_{kl}^I)}{2} \cdot \rho(i) \cdot \rho(j) \cdot \rho(k) \cdot \rho(l) \\ &\forall i = 1, \dots, n_E - 1, j = i + 1, \dots, n_E, k = 1, \dots, n_I - 1, l = k + 1, \dots, n_I \end{aligned} \quad (17)$$

$$1 - \sum_{p=1}^{n_I-1} \sum_{q=p+1}^{n_I} y_{ij}^{pq} = y_{ij}^E \quad \forall i = 1, \dots, n_E - 1, j = i + 1, \dots, n_E \quad (18)$$

$$1 - \sum_{p=1}^{n_E-1} \sum_{q=p+1}^{n_E} y_{pq}^{kl} = y_{kl}^I \quad \forall k = 1, \dots, n_I - 1, l = k + 1, \dots, n_I \quad (19)$$

$$\sum_{t=1}^T \sum_{s=1}^S y_{ijst}^E = y_{ij}^E \quad \forall i = 1, \dots, n_E - 1, j = i + 1, \dots, n_E \quad (20)$$

$$\sum_{t=1}^T \sum_{s=1}^S y_{klst}^I = y_{kl}^I \quad \forall k = 1, \dots, n_I - 1, l = k + 1, \dots, n_I \quad (21)$$

$$\begin{aligned} \sum_{t=1}^T \sum_{s=1}^S y_{ijst}^{kl} &= y_{ijl}^{kl} \\ &\forall i = 1, \dots, n_E - 1, j = i + 1, \dots, n_E, k = 1, \dots, n_I - 1, l = k + 1, \dots, n_I \end{aligned} \quad (22)$$

$$\sum_{t=1}^T \sum_{s=1}^S I_s \cdot y_{ijst}^E - g(t) \leq \delta_{ij}^E \quad \forall i = 1, \dots, n_E - 1, j = i + 1, \dots, n_E \quad (23)$$

$$\sum_{t=1}^T \sum_{s=1}^S I_s \cdot y_{klst}^I - g(t) \leq \delta_{kl}^I \quad \forall k = 1, \dots, n_I - 1, l = k + 1, \dots, n_I \quad (24)$$

$$\sum_{t=1}^T \sum_{s=1}^S I_s \cdot y_{ijst}^{kl} - g(t) \leq \delta_{ij}^{kl} \quad \forall i = 1, \dots, n_E - 1, j = i + 1, \dots, n_E, k = 1, \dots, n_I - 1, l = k + 1, \dots, n_I \quad (25)$$

$$\sum_{t=1}^T \sum_{s=1}^S F_s \cdot y_{ijst}^E - g(t) > -\delta_{ij}^E \quad \forall i = 1, \dots, n_E - 1, j = i + 1, \dots, n_E \quad (26)$$

$$\sum_{t=1}^T \sum_{s=1}^S F_s \cdot y_{klst}^I - g(t) > -\delta_{kl}^I \quad \forall k = 1, \dots, n_I - 1, l = k + 1, \dots, n_I \quad (27)$$

$$\sum_{t=1}^T \sum_{s=1}^S F_s \cdot y_{ijst}^{kl} - g(t) > -\delta_{ij}^{kl} \quad \forall i = 1, \dots, n_E - 1, j = i + 1, \dots, n_E, k = 1, \dots, n_I - 1, l = k + 1, \dots, n_I \quad (28)$$

$$\delta_{ij}^E \leq \delta_{max} \quad \forall i = 1, \dots, n_E - 1, j = i + 1, \dots, n_E \quad (29)$$

$$\delta_{kl}^I \leq \delta_{max} \quad \forall k = 1, \dots, n_I - 1, l = k + 1, \dots, n_I \quad (30)$$

$$\delta_{ij}^{kl} \leq \delta_{max} \quad \forall i = 1, \dots, n_E - 1, j = i + 1, \dots, n_E, k = 1, \dots, n_I - 1, l = k + 1, \dots, n_I \quad (31)$$

$$\begin{aligned} \sum_{t=1}^T \left(\sum_{i=1}^{n_E-1} \sum_{j=i+1}^{n_E} y_{ijst}^E + \sum_{i=1}^{n_E-1} \sum_{j=i+1}^{n_E} y_{ijst}^E - \sum_{i=1}^{n_E} y_{iist}^E \right. \\ \left. + \sum_{k=1}^{n_I-1} \sum_{l=k+1}^{n_I} y_{klst}^I + \sum_{k=1}^{n_I-1} \sum_{l=k+1}^{n_I} y_{klst}^I - \sum_{k=1}^{n_I} y_{kkst}^I \right. \\ \left. + \sum_{i=1}^{n_E-1} \sum_{j=i+1}^{n_E} \sum_{k=1}^{n_I-1} \sum_{l=k+1}^{n_I} y_{ijst}^{kl} + \sum_{i=1}^{n_E-1} \sum_{j=i+1}^{n_E} \sum_{k=1}^{n_I-1} \sum_{l=k+1}^{n_I} y_{ijst}^{kl} + \right. \\ \left. + \sum_{i=1}^{n_E-1} \sum_{j=i+1}^{n_E} \sum_{k=1}^{n_I-1} \sum_{l=k+1}^{n_I} y_{ijst}^{kl} + \sum_{i=1}^{n_E-1} \sum_{j=i+1}^{n_E} \sum_{k=1}^{n_I-1} \sum_{l=k+1}^{n_I} y_{ijst}^{kl} \right. \\ \left. - \sum_{i=1}^{n_E} \sum_{k=1}^{n_I-1} \sum_{l=k+1}^{n_I} y_{iist}^{kl} - \sum_{i=1}^{n_E-1} \sum_{j=i+1}^{n_E} \sum_{k=1}^{n_I} \sum_{l=k+1}^{n_I} y_{ijst}^{kl} - \sum_{i=1}^{n_E} \sum_{k=1}^{n_I} \sum_{l=k+1}^{n_I} y_{iist}^{kl} \right) \leq Q_s^a \\ \forall a = 1, \dots, A, s = 1, \dots, S \quad (32) \end{aligned}$$

$$x_{ij}^E \in \{0, 1\} \quad \forall i = 1, \dots, n_E, j = 1, \dots, n_E \quad (33)$$

$$x_{kl}^I \in \{0, 1\} \quad \forall k = 1, \dots, n_I, l = 1, \dots, n_I \quad (34)$$

$$y_{ij}^{kl}, y_{ijst}^{kl} \in \{0, 1\} \quad \forall i = 1, \dots, n_E, j = 1, \dots, n_E, k = 1, \dots, n_I, l = 1, \dots, n_I \quad (35)$$

$$y_{ij}^E, y_{ijst}^E \in \{0, 1\} \quad \forall i = 1, \dots, n_E, j = 1, \dots, n_E \quad (36)$$

$$y_{kl}^I, y_{klst}^I \in \{0, 1\} \quad \forall k = 1, \dots, n_I, l = 1, \dots, n_I \quad (37)$$

$$\delta_{ij}^E, \delta_{kl}^I, \delta_{ij}^{kl} \geq 0 \quad \forall i = 1, \dots, n_E, j = 1, \dots, n_E, k = 1, \dots, n_I, l = 1, \dots, n_I \quad (38)$$

The first and second terms of the objective function (12) favor the formation of export and import tuples, respectively, with minimum cardinality. The third term of the objective function motivates the creation of complete tuples. The remaining terms of the objective function minimize the deviation of trucks from their preferred time slot.

Constraints (13)–(19) are required to construct the tuples. Constraints (13) and (14) are used to group export containers i and j in pairs if $j \in U(i)$. Constraints (15) and (16) are used for the same purpose, but for import containers. Constraints (17) are used to form complete tuples; they state that if export and import containers have a collaboration agreement, they can form complete tuples. Constraints (18) establish that if export containers i and j did not match with import containers, the truck will return in an empty trip. Constraints (19) establishes that if import containers k and l did not match with export containers, the truck will produce an empty trip. Constraints (20)–(22) are used to ensure that each one of the eight types of tuples are assigned to exactly one time slot. Constraints (23)–(28) are used to compute the deviation from the preferred time slot for each type of tuple. Constraints (29)–(31) set a maximum limit for this deviation. Constraints (32) set a capacity condition establishing that in area a at most Q_s^a moves may be performed during time slot s . Finally, constraints (33)–(38) define the nature of the decision variables.

It can be noted that, in its essence, Model 2 is a combination of a *matching problem* and a *scheduling problem*, which are NP-hard.

The formation of export and import tuples is modeled as a minimum maximal matching problem, while the assignment of tuples to time slots aiming to minimize deviation is a scheduling problem with capacity constraints. This formulation requires $O(E^2 + I^2 + (1+s)(E^2 + I^2 + E^2I^2))$ binary variables and $O(6E^2 + 6I^2 + 5E^2I^2 + E + I + as)$ constraints, where E and I are the number of export and import containers, respectively.

Appendix B. Rules to determine the preferred time slot

The preferred time slot of tuple n (g_n) is determined depending on the type of tuple identified in Table 2 (case 1 to 8), according to the following procedure:

- case 1: the tuple deadline, i.e. the preferred time slot, has to respect the vessel departure time (attribute vi) of the export container. This means that the truck assigned to this tuple must arrive to the port in the yard opening time (which indicatively starts 1 week in advance respect to its vessel departure and usually closes some hours before the customs closure);
- case 2: the tuple deadline has to respect the vessel departure times of the two export containers (obviously to be able to match two export containers, their yard opening times must overlap);
- case 3: the truck is assigned to one import container considering its dwell time (attribute ix), trying to avoid to overcome the number of free charge days allowed by the terminal;
- case 4: the truck is assigned to two import containers considering their dwell times, trying to avoid to overcome the number of free charge days allowed by the terminal for both of them (obviously upstream, only import containers that have similar dwell times will be considered to form tuples);
- case 5: the truck is assigned to one import and one export containers matched together, giving priority to the vessel departure time for the export container and, secondly, trying not to overcome the number of free charge days for the import container (obviously upstream, only import and export containers having close deadlines will be considered to form tuples);
- case 6: the truck is assigned to two exports and one import containers matched together, giving priority to the vessel departure times for the export containers and, secondly, trying not to overcome the number of free charge days for the import container;
- case 7: the truck is assigned to one export and two import containers matched together, giving priority to the vessel departure time for the export container and, secondly, trying not to overcome the number of free charge days for the two import containers;
- case 8: the truck is assigned to two export and two import containers matched together, giving priority to the vessel departing times for the export containers and, secondly, trying not to overcome the number of free charge days for the import containers.

Appendix C. The relation between Factor F2 and the attribute “agreement”

The attribute “agreement”, which is related to each individual container, is a categorical variable that captures the existence of collaboration agreements between shippers or freight forwarders to share their trucks in single or double cycles. All the containers belonging to shippers/freight forwarders having an agreement share a common code and may be transported in the same tuple/truck.

In order to explain how Factor F2 is associated with the attribute “agreement”, let us consider the following two cases with 15 containers, eight export and seven import. In both cases, the containers that share a common colour have an agreement to collaborate and may be transported in the same truck or, in other words, may be assigned to the same tuple; on the other hand, the containers without colour (the white ones) have not an agreement and must be transported alone.

In case A (Fig. 17), 11 containers (11/15 $\approx$ 73.33%) have an agreement and the remaining 4 containers do not. Therefore, if the other criteria used to form tuples allow it, the following eight tuples may be formed:

- Tuple1 = [E1, E2, I11, -]
- Tuple2 = [E3, -, I9, -]
- Tuple3 = [E4, -, -, -]
- Tuple4 = [E5, E7, I10, I15]
- Tuple5 = [E6, -, -, -]
- Tuple6 = [E8, -, -, -]
- Tuple7 = [-, -, I13, I14]
- Tuple8 = [-, -, I12, -]

Note that three tuples (1, 2, and 4) perform a double move, whereas Tuple 7 moves two containers but in a single move. The portion of tuples performing double moves is $3/8 = 37.5\%$.

In case B (Fig. 18), only 6 containers (6/15 $\approx$ 40%) have an agreement and the remaining 9 containers do not. Therefore, if the other criteria used to form tuples allow it, the following twelve tuples may be formed:



Fig. 17. Presence of container agreement attributes - Case A.

E1	E2	E3	E4	E5	E6	E7	E8	I9	I10	I11	I12	I13	I14	I15
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Fig. 18. Presence of container agreement attributes - Case B.

- Tuple1 = [E1, -, I12, -]
- Tuple2 = [E2, -, -, -]
- Tuple3 = [E3, -, I9, -]
- Tuple4 = [E4, -, -, -]
- Tuple5 = [E5, -, I10, -]
- Tuple6 = [E6, -, -, -]
- Tuple7 = [E7, -, -, -]
- Tuple8 = [E8, -, -, -]
- Tuple9 = [-, -, I11, -]
- Tuple10 = [-, -, I13, -]
- Tuple11 = [-, -, I14, -]
- Tuple12 = [-, -, I15, -]

In this case, three tuples (1, 3, and 5) perform a double move but the portion of tuples performing double moves is lower, i.e., $3/12 = 25.0\%$.

Note that, in Case A, 73% of the containers have an agreement, which means that the remaining 27% are transported alone in a single trip, so generating an empty trip each. In Case B, where the percentage of containers having an agreement is much lower (i.e., 40%), a higher number (60%) of empty trips is performed.

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